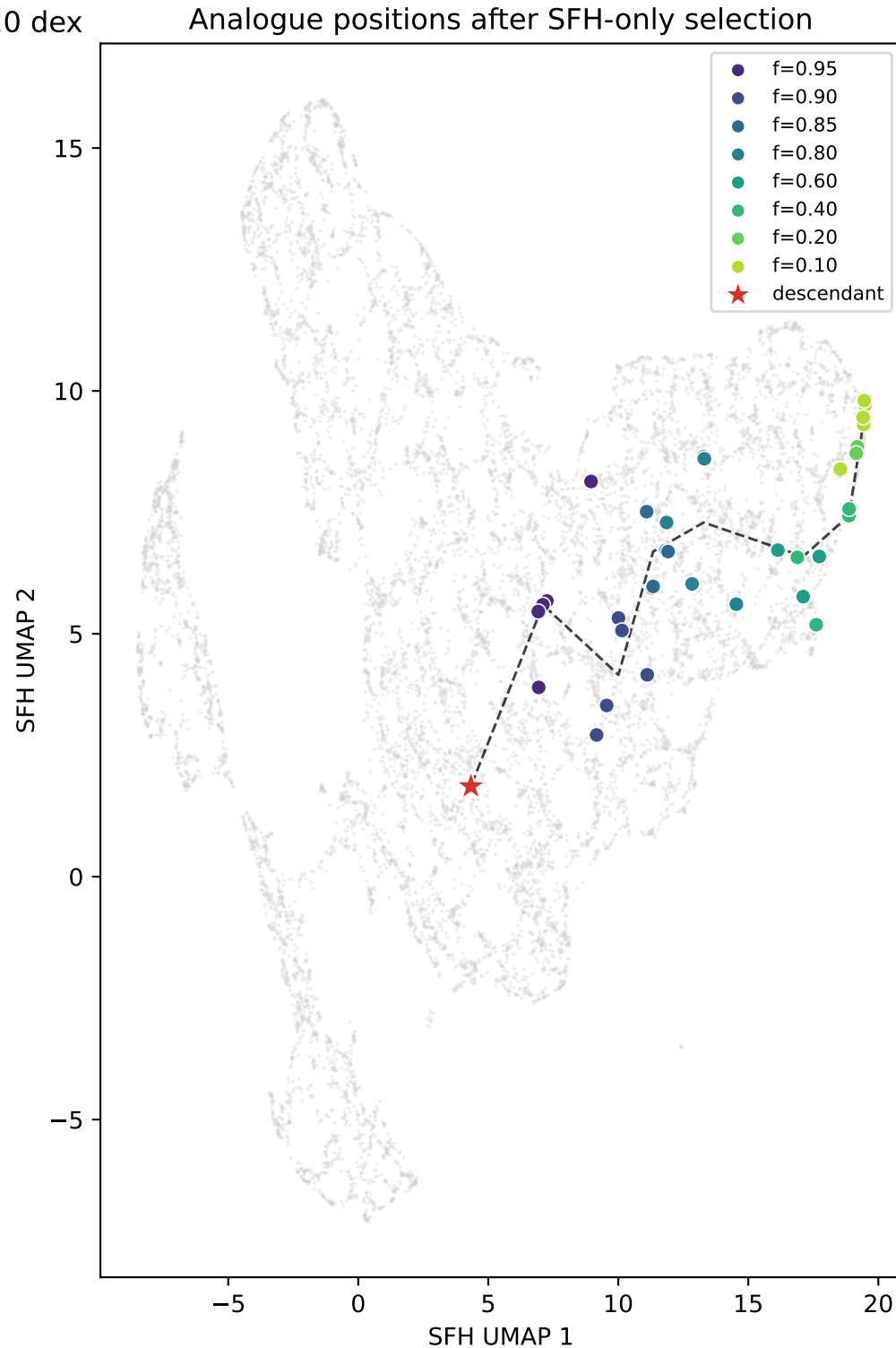
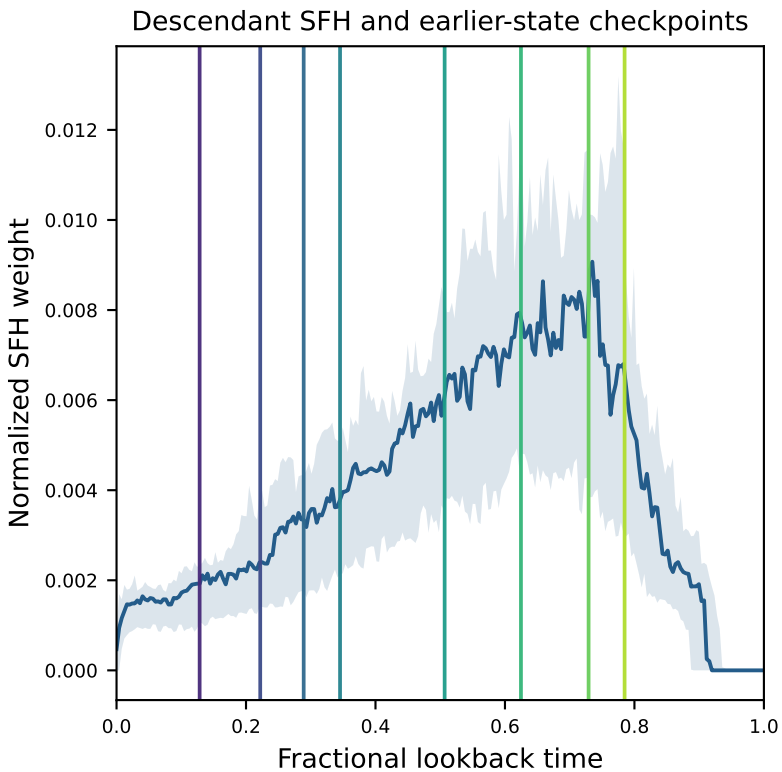
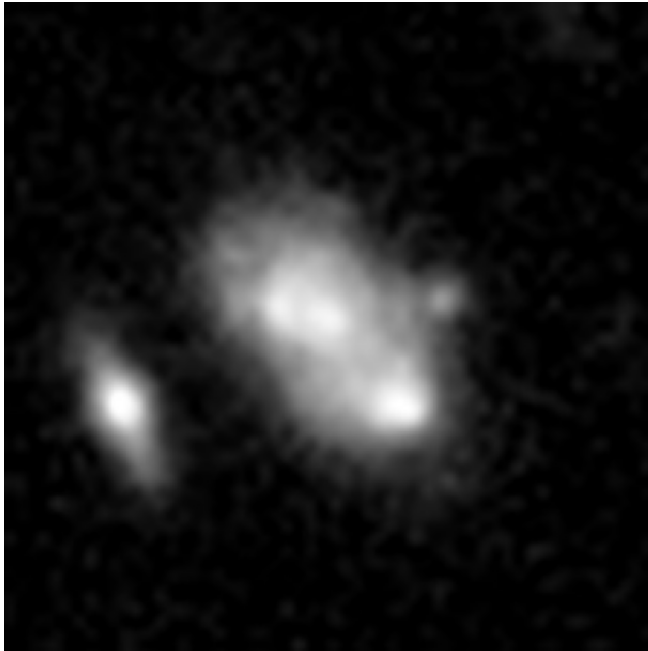
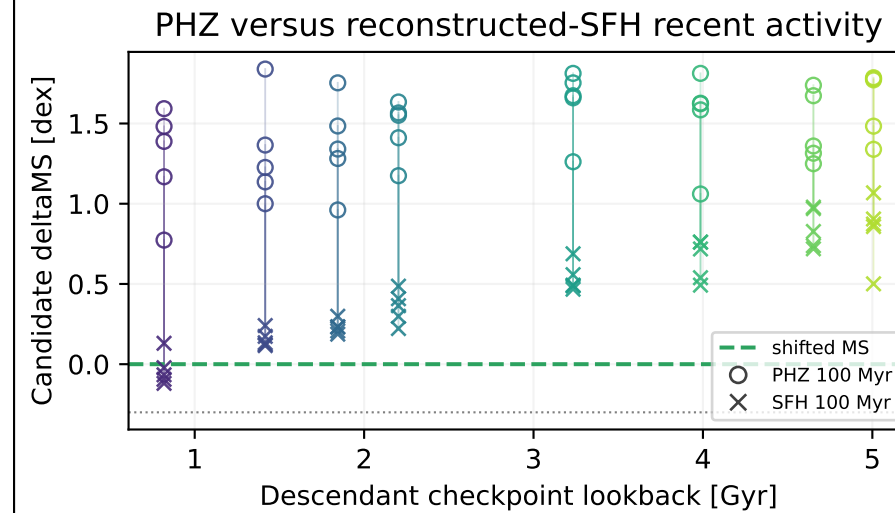


Pilot SFH-selected progenitor-analogue sequence

Descendant 2701535932638430142
 $\log M^*=10.48$, $z=0.854$, $\log \text{sSFR}=-9.17$
 PHZ $\text{deltaMS}=+0.98$; SFH-100Myr $\text{deltaMS}=-0.20$ dex



f	L	logM	T	N	zmed
0.95	0.82	10.46	2.00	1,952	0.57
0.90	1.42	10.44	2.00	1,960	0.57
0.85	1.84	10.41	2.00	1,961	0.56
0.80	2.20	10.39	2.00	1,963	0.56
0.60	3.23	10.26	2.00	2,009	0.54
0.40	3.99	10.09	2.00	2,047	0.50
0.20	4.65	9.79	1.73	1,914	0.42
0.10	5.01	9.48	1.37	1,661	0.35



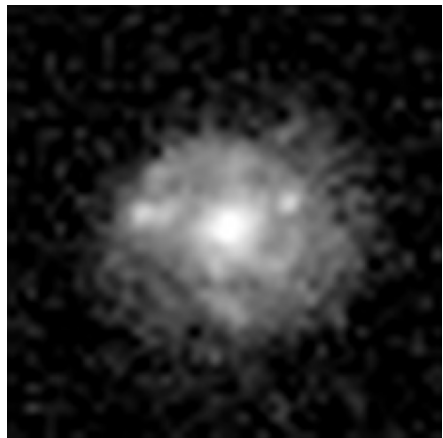
Selection variables

- predicted stellar mass, within ± 0.15 dex
- cumulative SFH before each descendant state
- 2 Gyr physical-time comparison when available

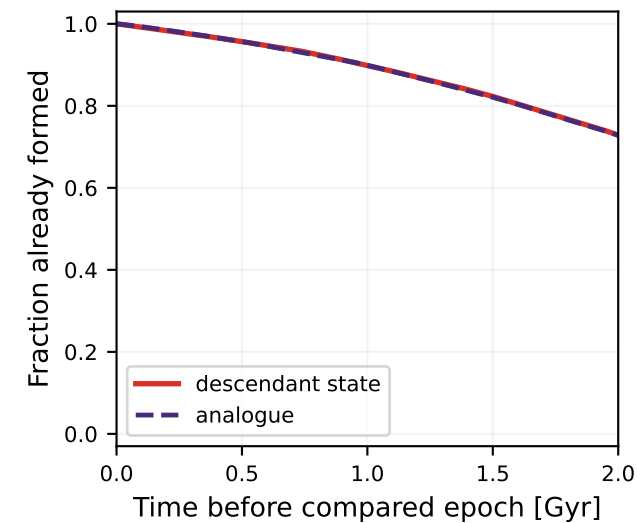
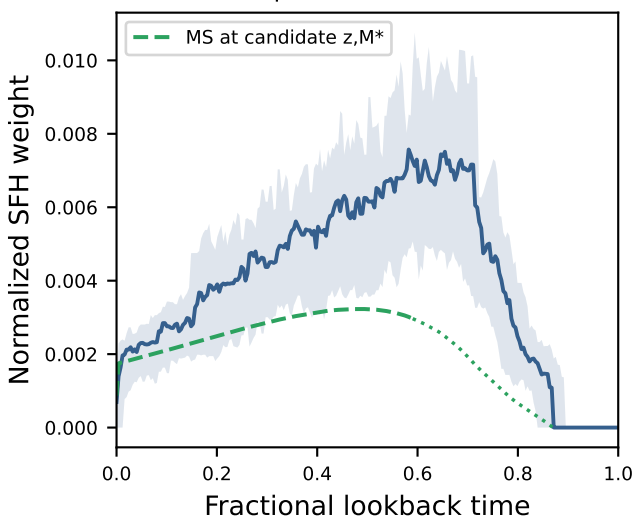
Candidate redshift, morphology, deltaMS , image embedding, and UMAP position are outcomes, not selection variables. The dashed path joins ensemble medians, not one galaxy's orbit.

Earlier state $f=0.95$: lookback=0.82 Gyr, predicted $\log M^*=10.46$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

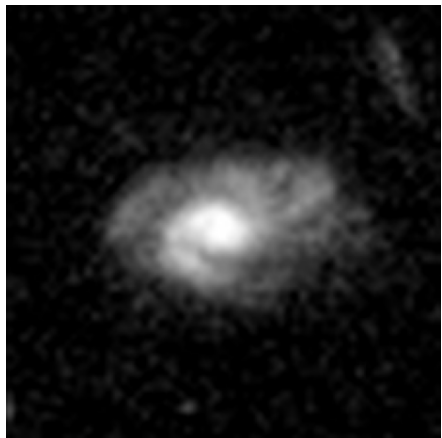
rank 1: 2765823515651281065
 $\log M^*=10.61 (+0.15)$ $z=0.916$
 PHZ $\Delta MS=+1.39$; SFH-100Myr $\Delta MS=-0.02$
 $D=0.001$



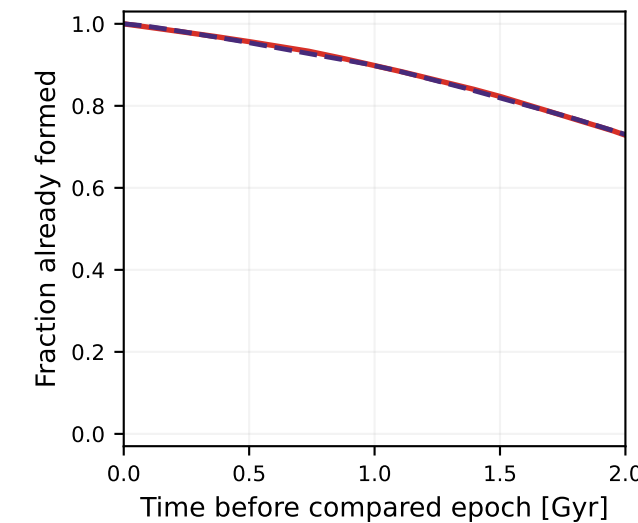
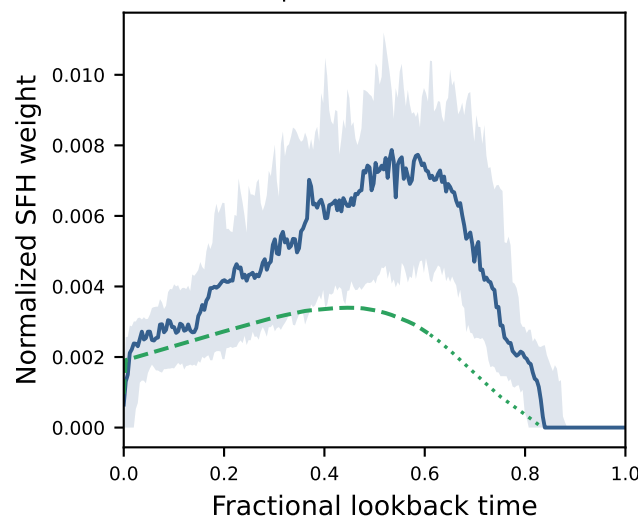
Complete candidate SFH



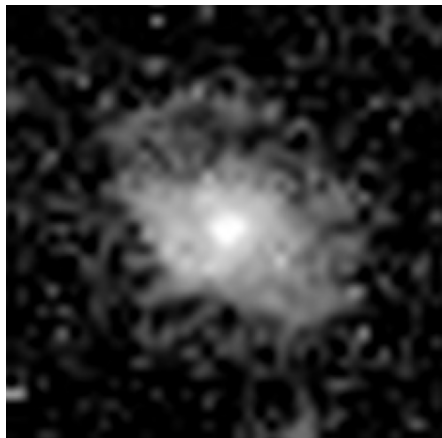
rank 2: 2715637870685853669
 $\log M^*=10.40 (-0.06)$ $z=0.848$
 PHZ $\Delta MS=+1.17$; SFH-100Myr $\Delta MS=-0.07$
 $D=0.002$



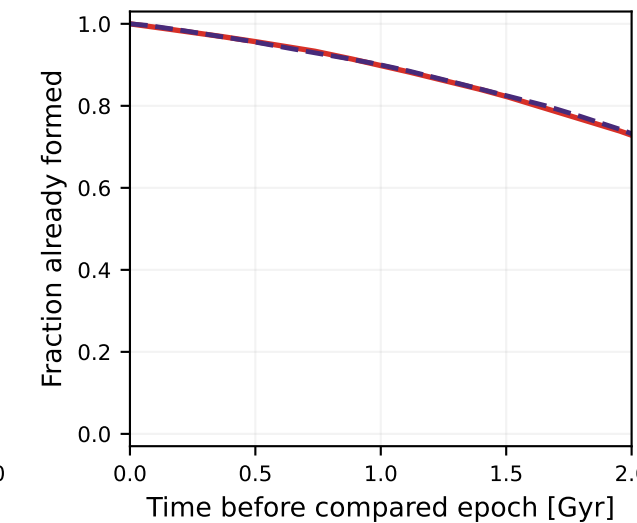
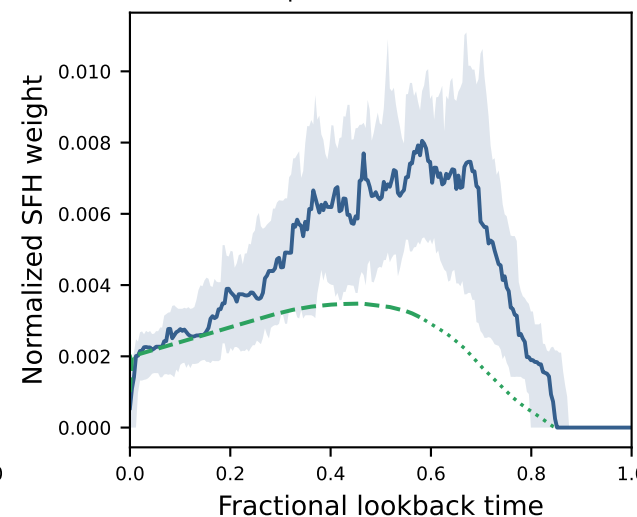
Complete candidate SFH



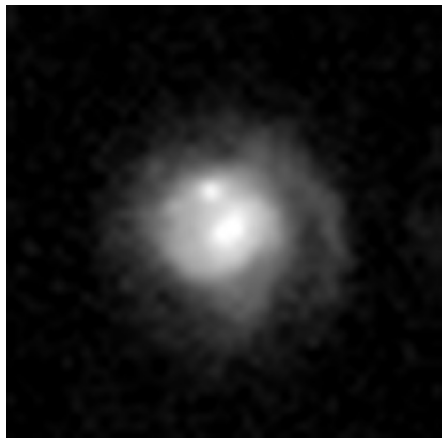
rank 3: 2758528497640400544
 $\log M^*=10.45 (-0.01)$ $z=0.934$
 PHZ $\Delta MS=+1.48$; SFH-100Myr $\Delta MS=-0.12$
 $D=0.002$



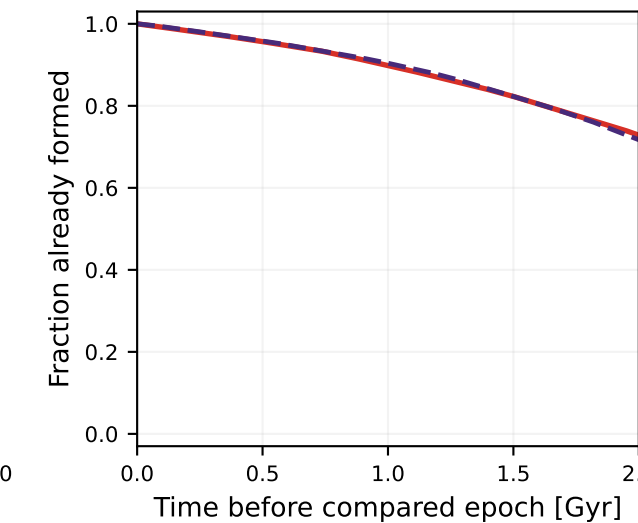
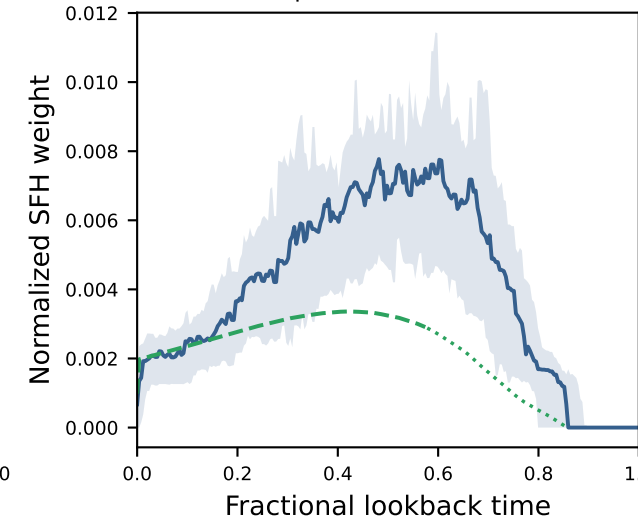
Complete candidate SFH



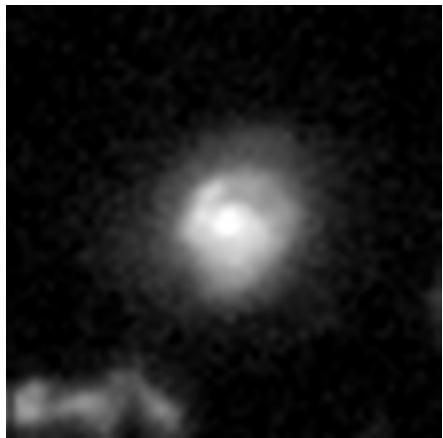
rank 4: 2686438568655226633
 $\log M^*=10.50 (+0.04)$ $z=0.962$
 PHZ $\Delta MS=+1.59$; SFH-100Myr $\Delta MS=-0.09$
 $D=0.003$



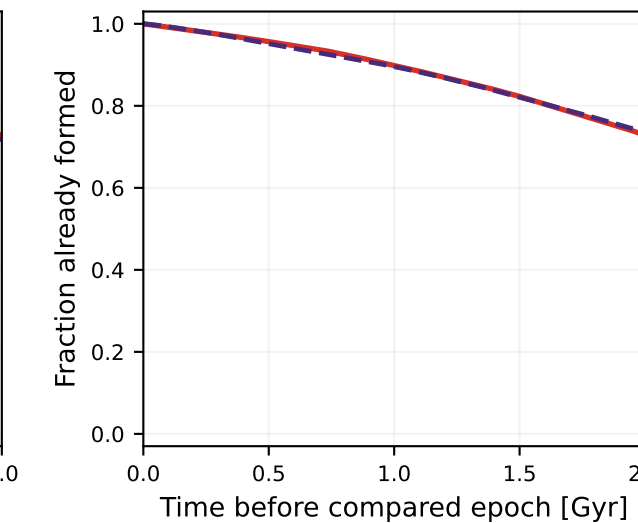
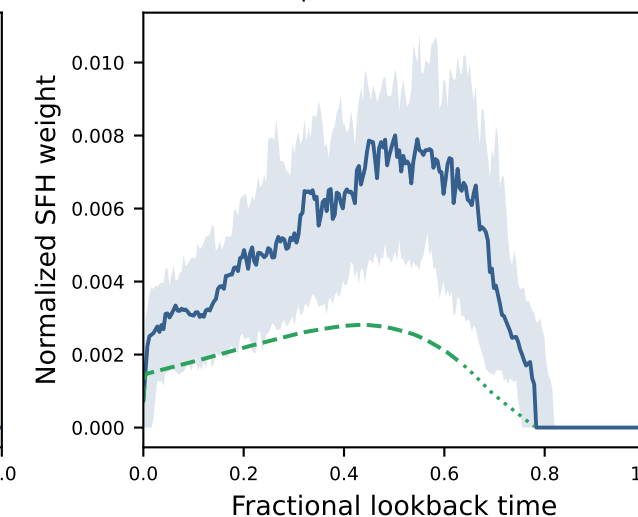
Complete candidate SFH



rank 5: 2654381422648556082
 $\log M^*=10.56 (+0.10)$ $z=0.720$
 PHZ $\Delta MS=+0.77$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.003$



Complete candidate SFH



Earlier state $f=0.90$: lookback=1.42 Gyr, predicted $\log M^*=10.44$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

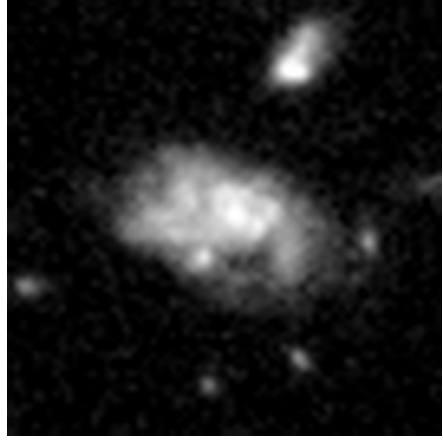
rank 1: 2665034043668981542
 $\log M^*=10.34$ (-0.10) $z=0.860$
 PHZ $\Delta MS=+1.00$; SFH-100Myr $\Delta MS=+0.12$
 $D=0.003$

rank 2: 2673574569652995115
 $\log M^*=10.47$ (+0.03) $z=0.990$
 PHZ $\Delta MS=+1.37$; SFH-100Myr $\Delta MS=+0.12$
 $D=0.004$

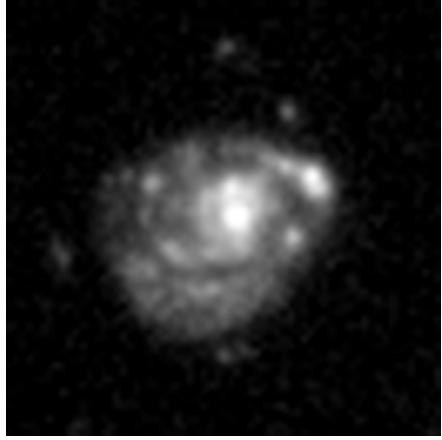
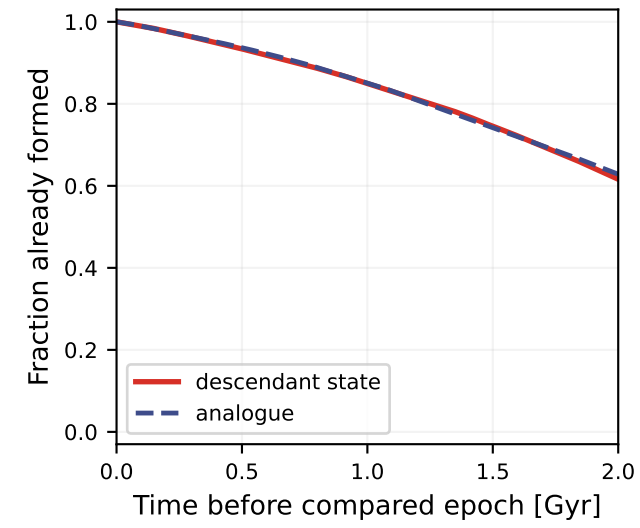
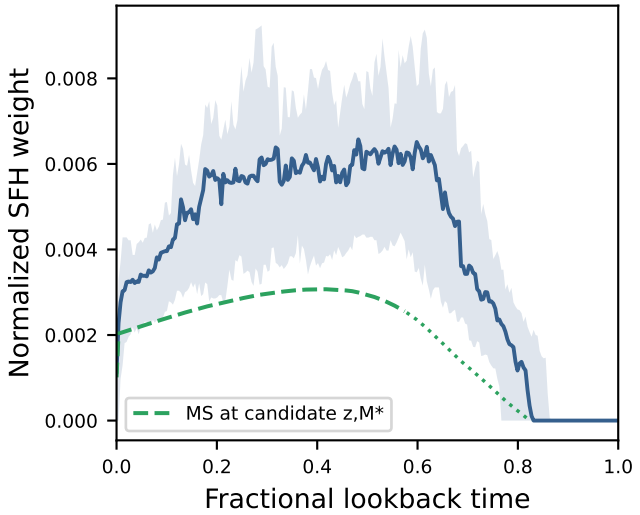
rank 3: 2643456408663786240
 $\log M^*=10.44$ (+0.01) $z=0.936$
 PHZ $\Delta MS=+1.23$; SFH-100Myr $\Delta MS=+0.13$
 $D=0.004$

rank 4: 2727579410637599231
 $\log M^*=10.29$ (-0.15) $z=0.726$
 PHZ $\Delta MS=+1.14$; SFH-100Myr $\Delta MS=+0.24$
 $D=0.006$

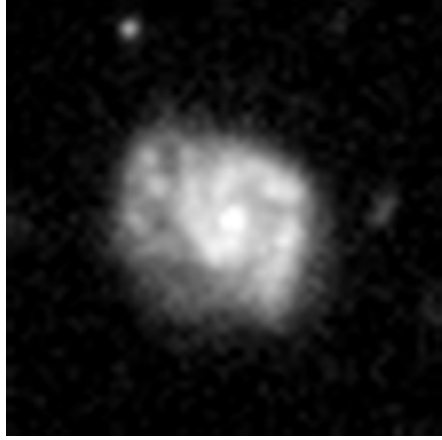
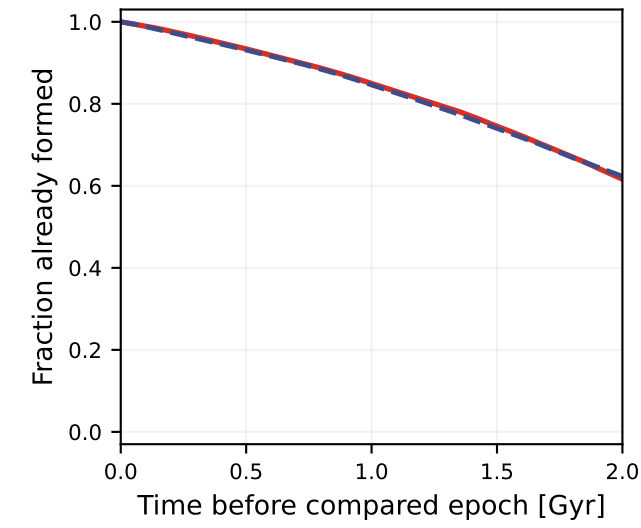
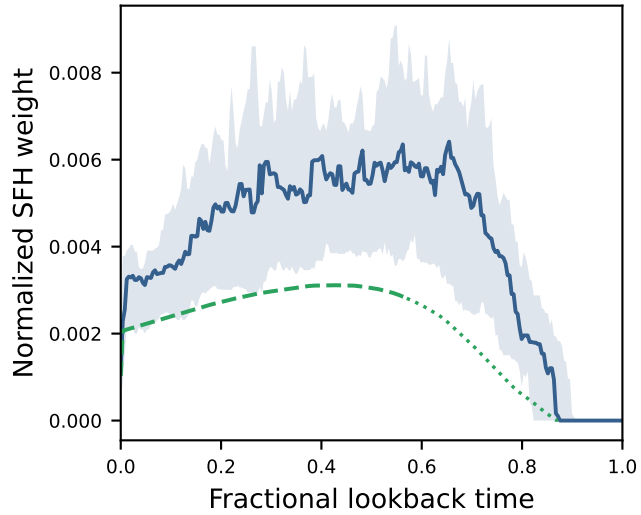
rank 5: 2746076314657644554
 $\log M^*=10.45$ (+0.01) $z=0.930$
 PHZ $\Delta MS=+1.84$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.006$



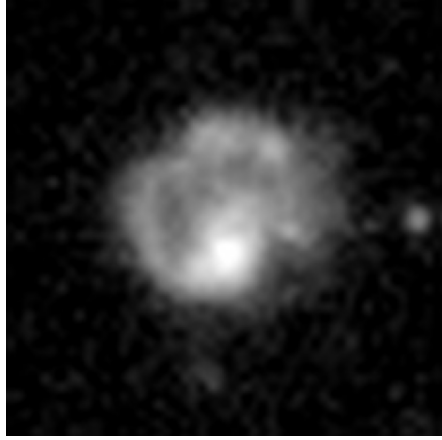
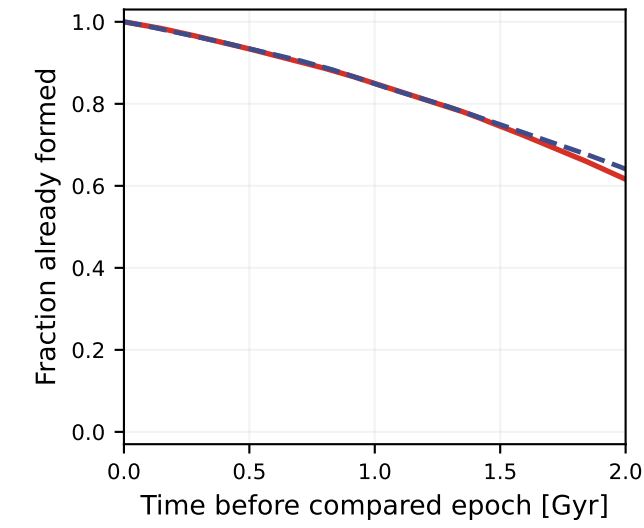
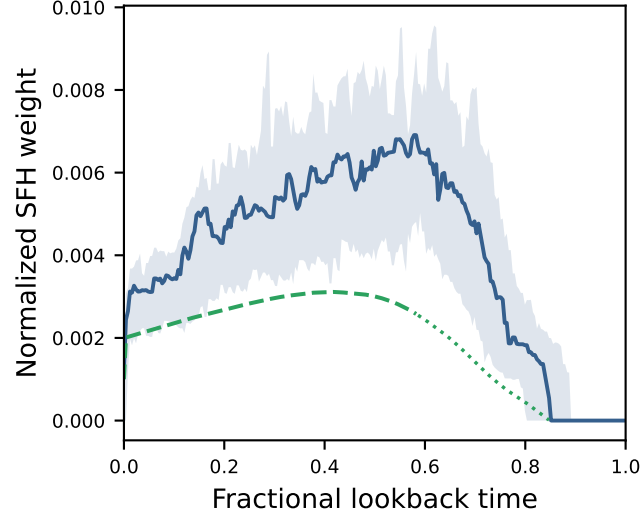
Complete candidate SFH



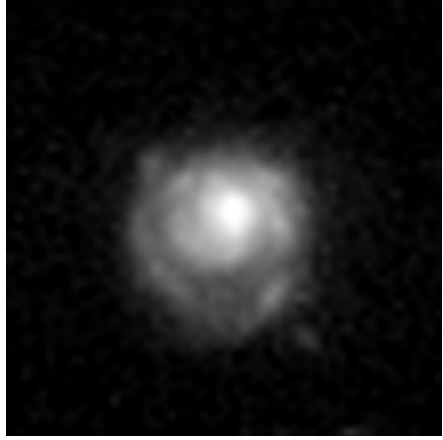
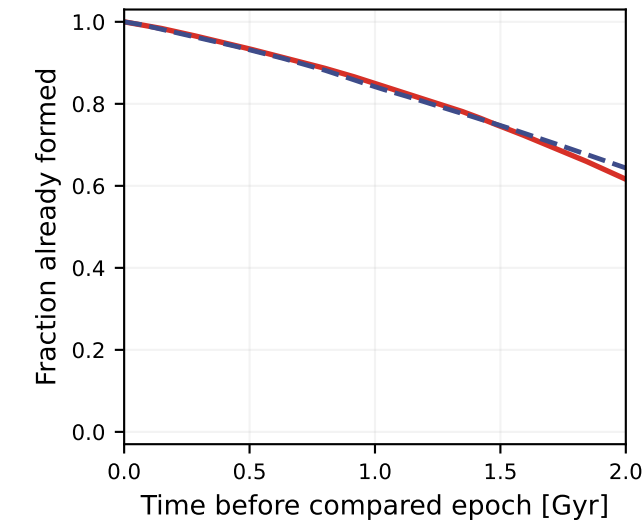
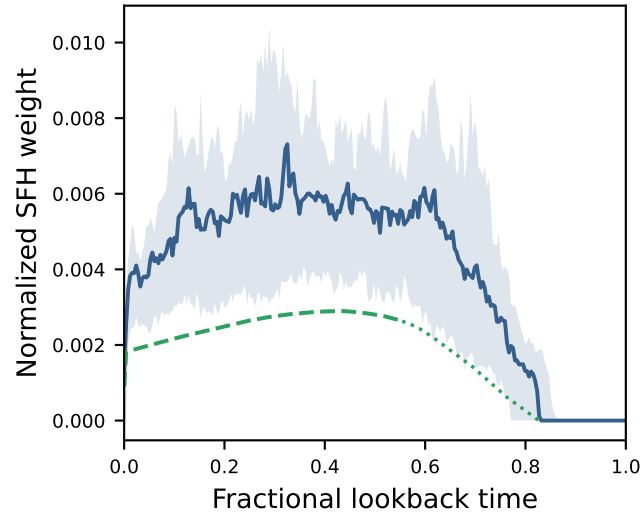
Complete candidate SFH



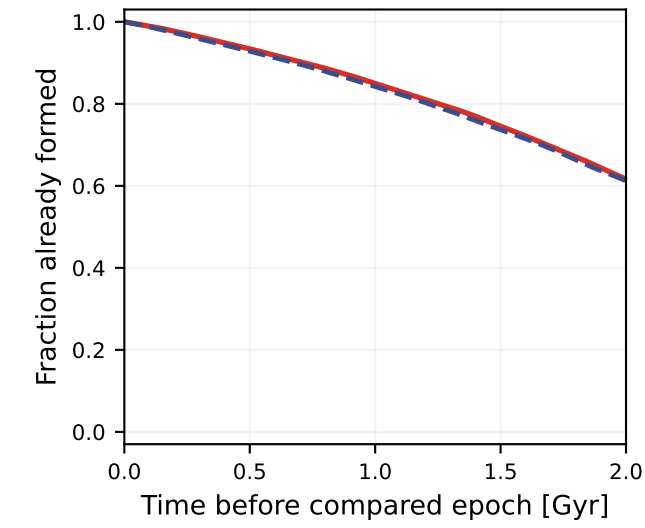
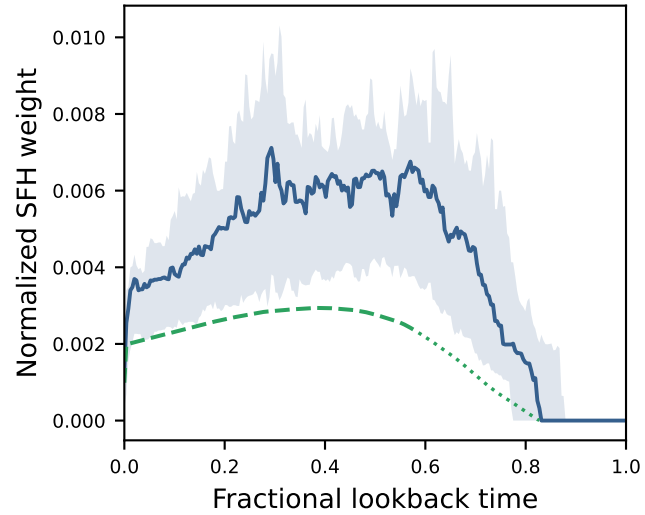
Complete candidate SFH



Complete candidate SFH

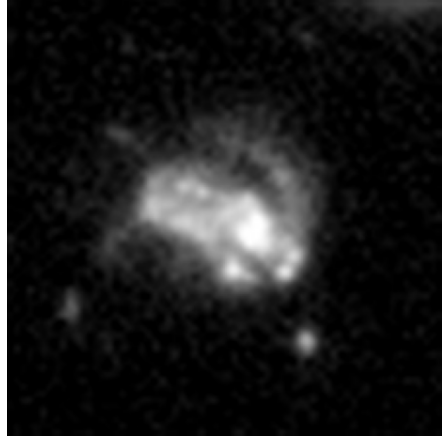


Complete candidate SFH

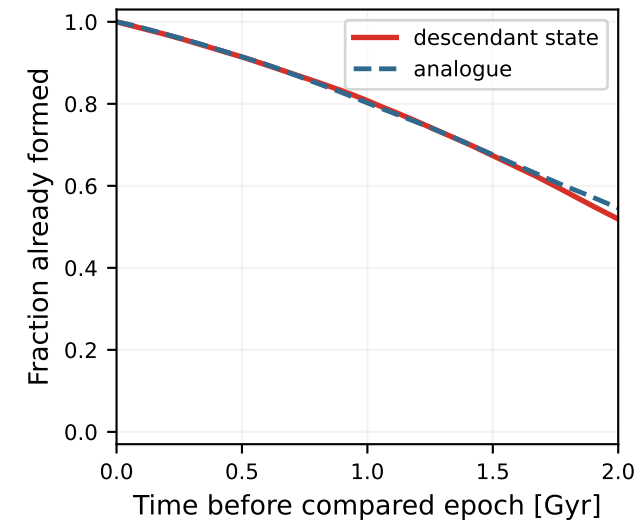
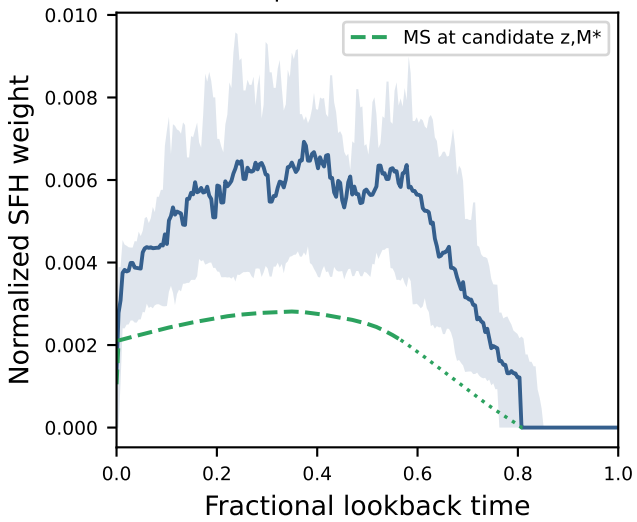


Earlier state $f=0.85$: lookback=1.84 Gyr, predicted $\log M^*=10.41$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

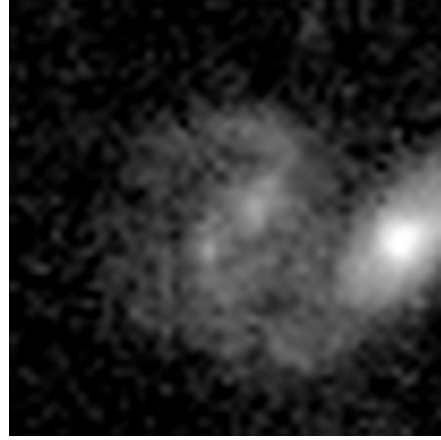
rank 1: 2729826866681253754
 $\log M^*=10.44$ (+0.03) $z=0.992$
 PHZ $\Delta MS=+1.75$; SFH-100Myr $\Delta MS=+0.19$
 $D=0.004$



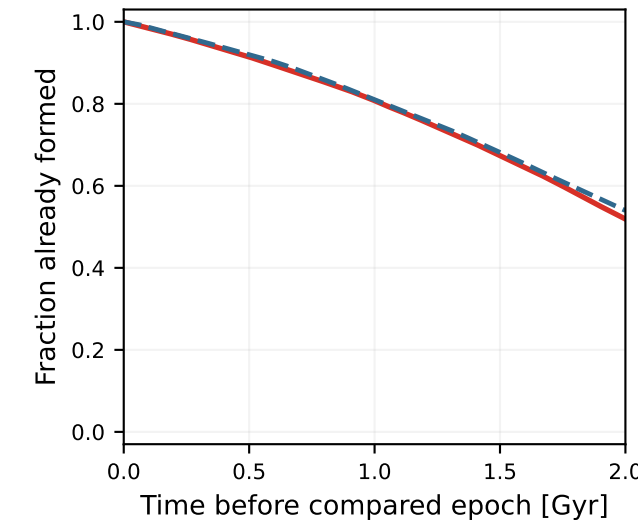
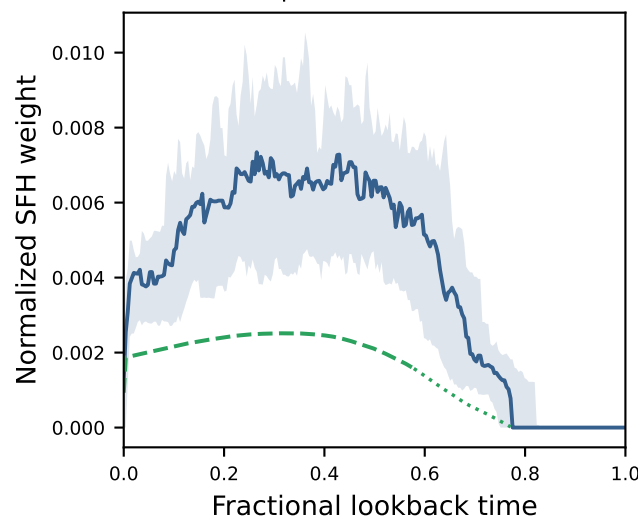
Complete candidate SFH



rank 2: 2625587373653760688
 $\log M^*=10.54$ (+0.12) $z=0.932$
 PHZ $\Delta MS=+1.28$; SFH-100Myr $\Delta MS=+0.23$
 $D=0.007$



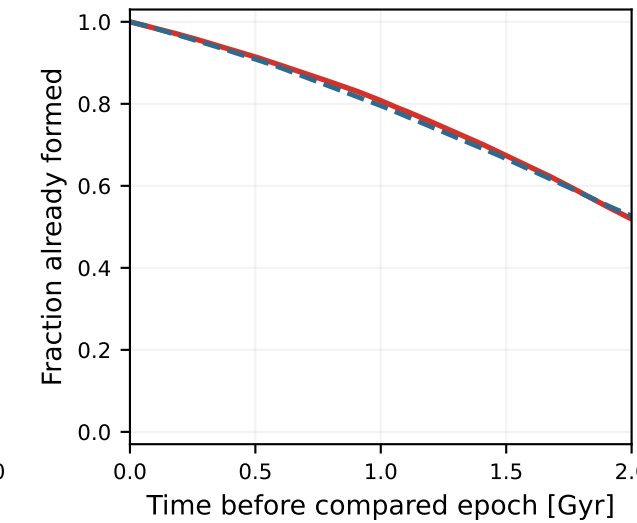
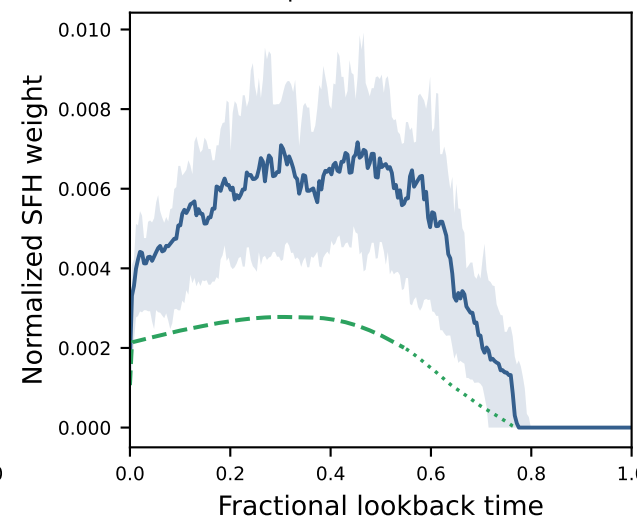
Complete candidate SFH



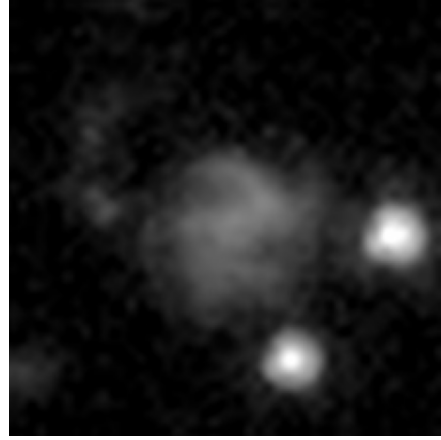
rank 3: 2669962211655798316
 $\log M^*=10.41$ (-0.01) $z=0.978$
 PHZ $\Delta MS=+1.48$; SFH-100Myr $\Delta MS=+0.23$
 $D=0.007$



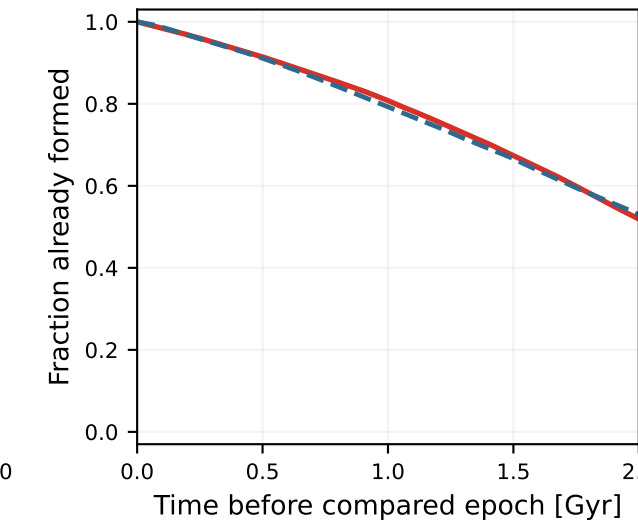
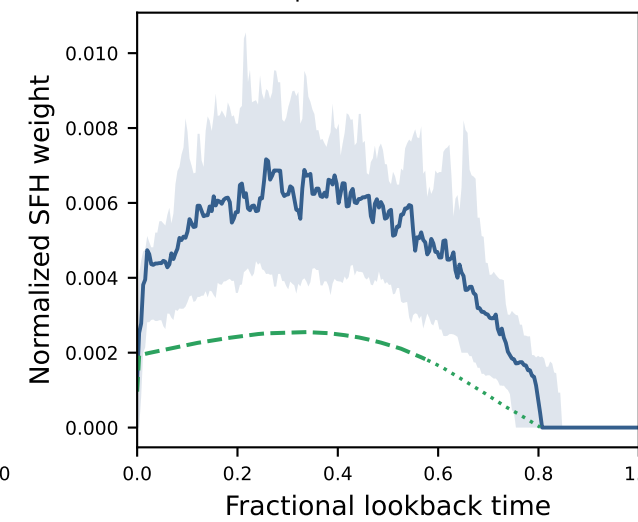
Complete candidate SFH



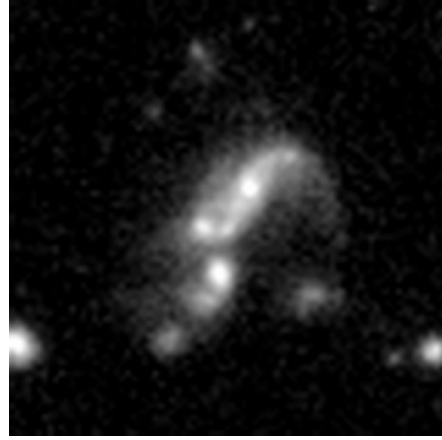
rank 4: 2757534665642581103
 $\log M^*=10.53$ (+0.11) $z=0.960$
 PHZ $\Delta MS=+0.96$; SFH-100Myr $\Delta MS=+0.21$
 $D=0.007$



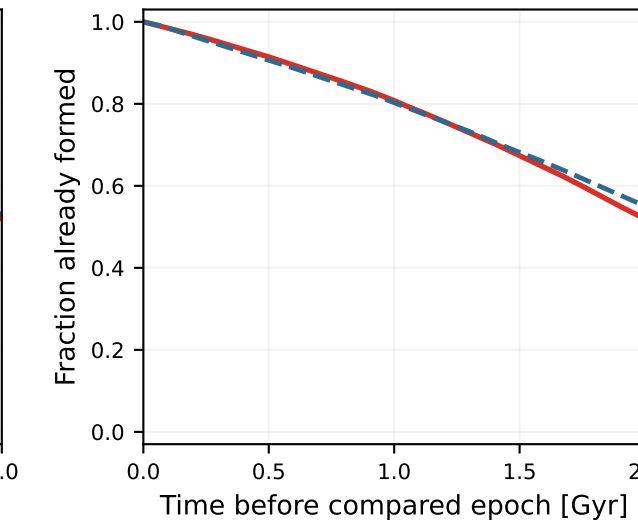
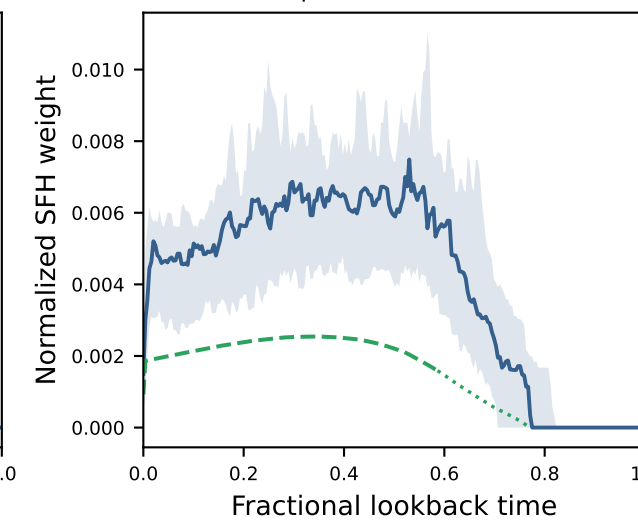
Complete candidate SFH



rank 5: 2659004518638398653
 $\log M^*=10.54$ (+0.13) $z=0.930$
 PHZ $\Delta MS=+1.34$; SFH-100Myr $\Delta MS=+0.21$
 $D=0.009$

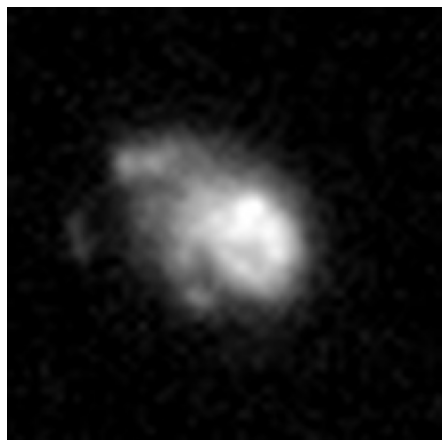


Complete candidate SFH

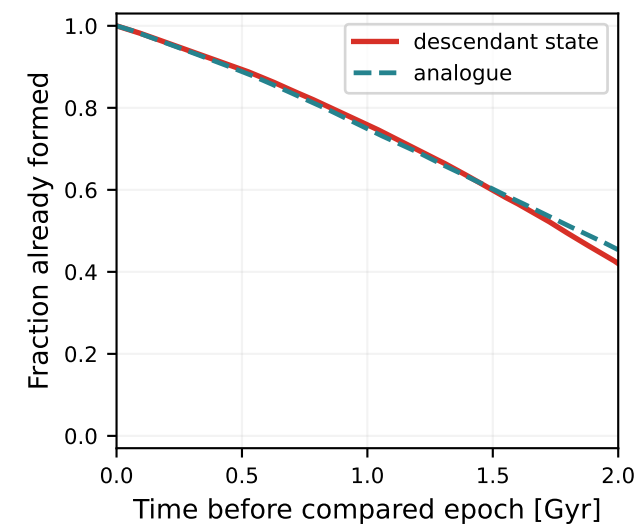
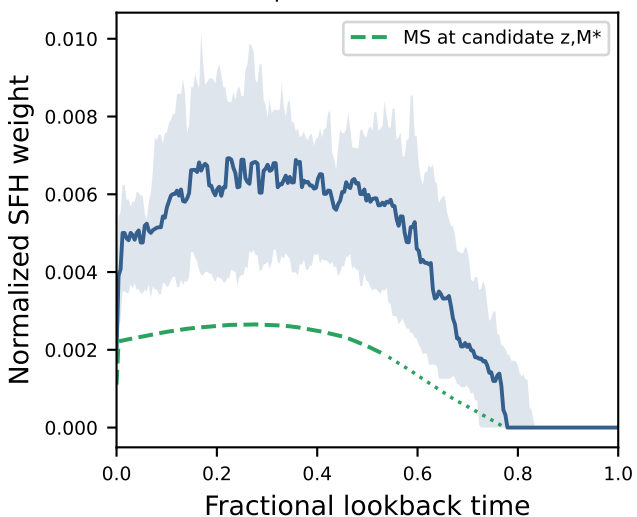


Earlier state $f=0.80$: lookback=2.20 Gyr, predicted $\log M^*=10.39$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

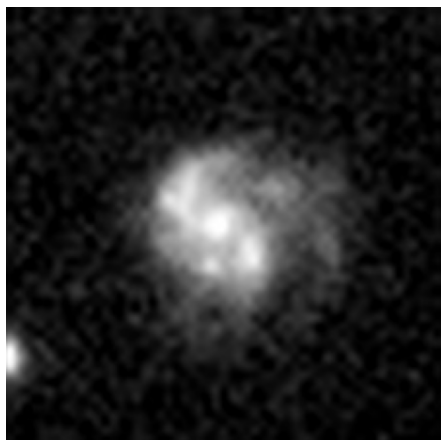
rank 1: 2747650982670270769
 $\log M^*=10.45$ (+0.07) $z=1.072$
 PHZ $\Delta MS=+1.41$; SFH-100Myr $\Delta MS=+0.30$
 $D=0.008$



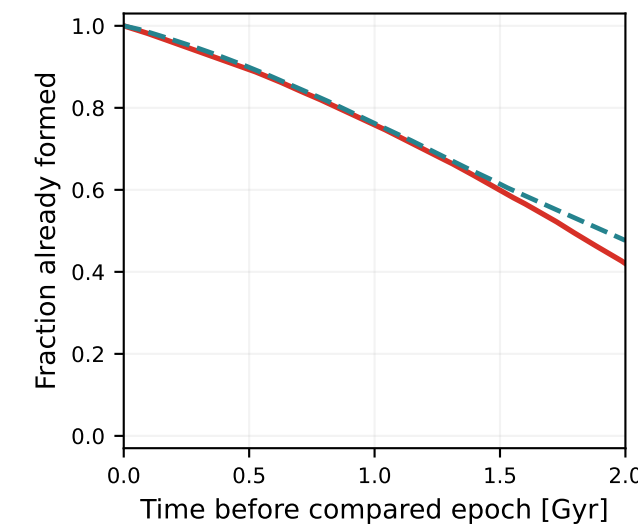
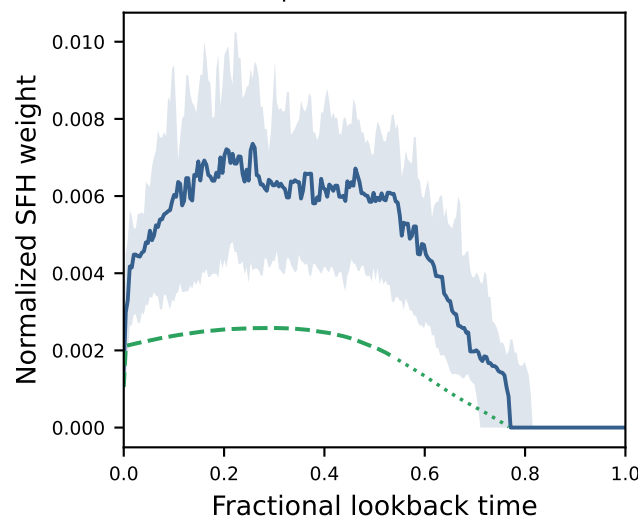
Complete candidate SFH



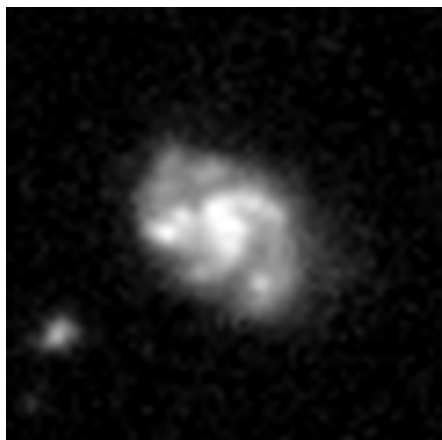
rank 2: 2650202959678670856
 $\log M^*=10.45$ (+0.06) $z=1.006$
 PHZ $\Delta MS=+1.55$; SFH-100Myr $\Delta MS=+0.22$
 $D=0.013$



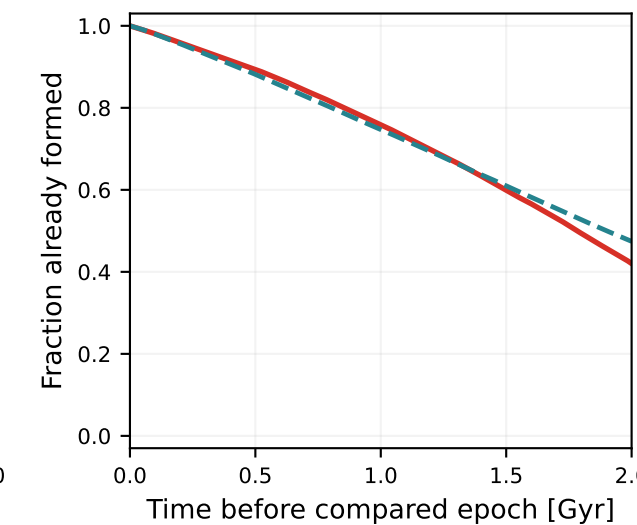
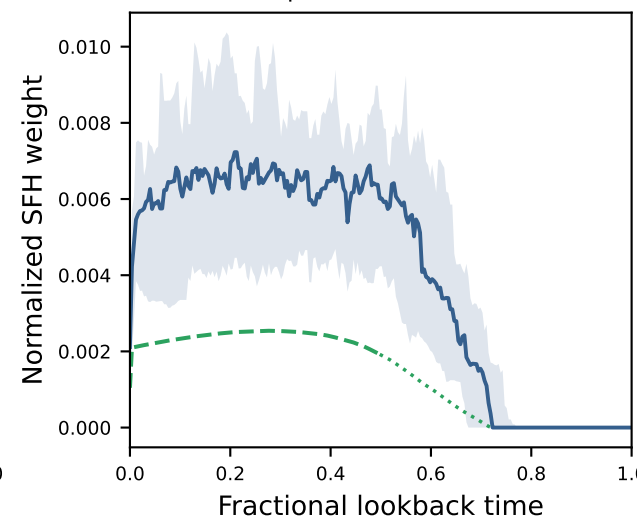
Complete candidate SFH



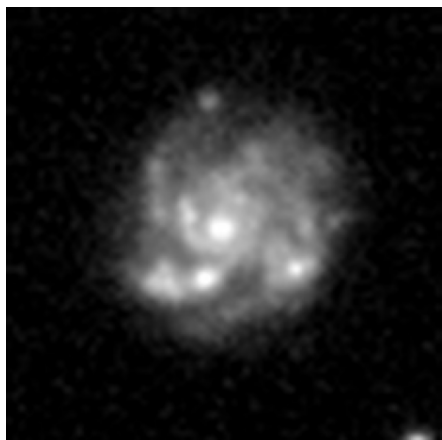
rank 3: 2658537943673551922
 $\log M^*=10.37$ (-0.01) $z=0.928$
 PHZ $\Delta MS=+1.57$; SFH-100Myr $\Delta MS=+0.36$
 $D=0.013$



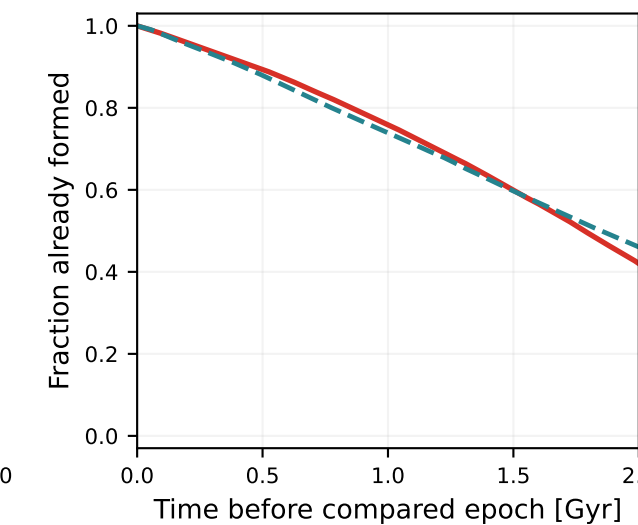
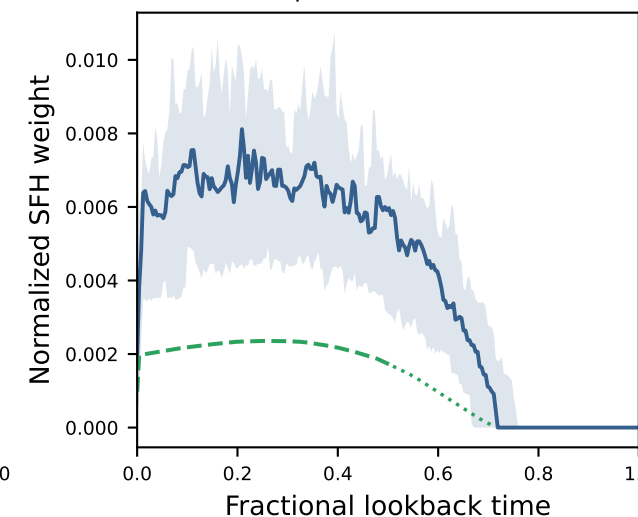
Complete candidate SFH



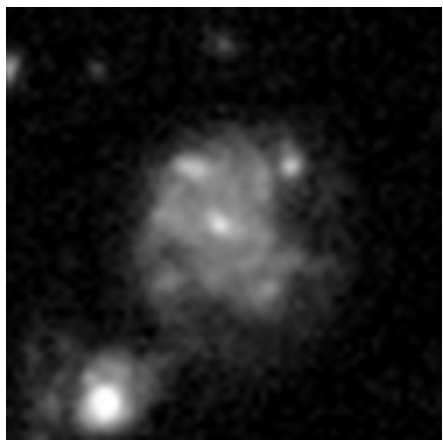
rank 4: 2740149368667834322
 $\log M^*=10.45$ (+0.06) $z=0.918$
 PHZ $\Delta MS=+1.63$; SFH-100Myr $\Delta MS=+0.41$
 $D=0.014$



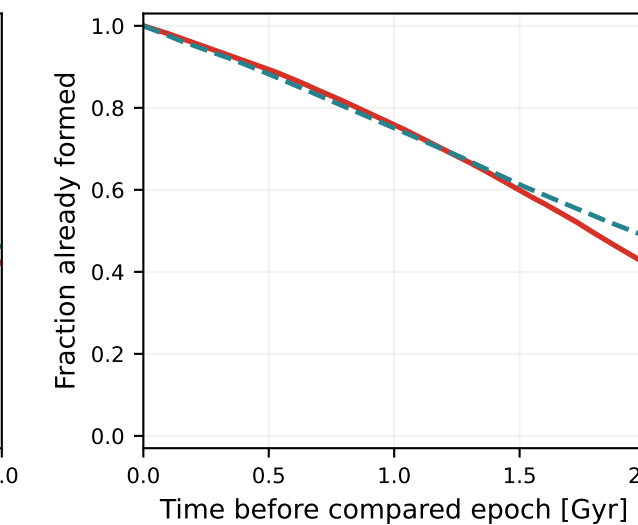
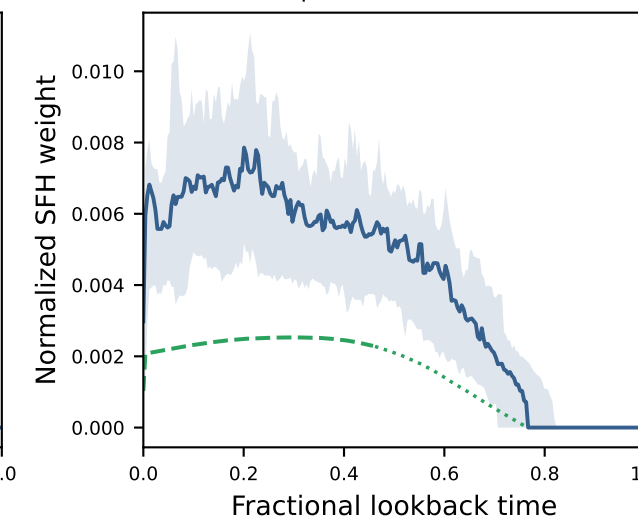
Complete candidate SFH



rank 5: 2719975428650465420
 $\log M^*=10.27$ (-0.12) $z=0.834$
 PHZ $\Delta MS=+1.17$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.015$

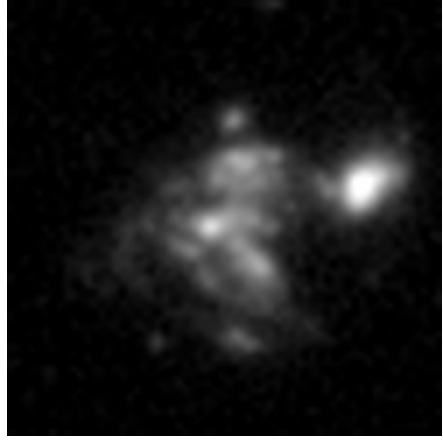


Complete candidate SFH

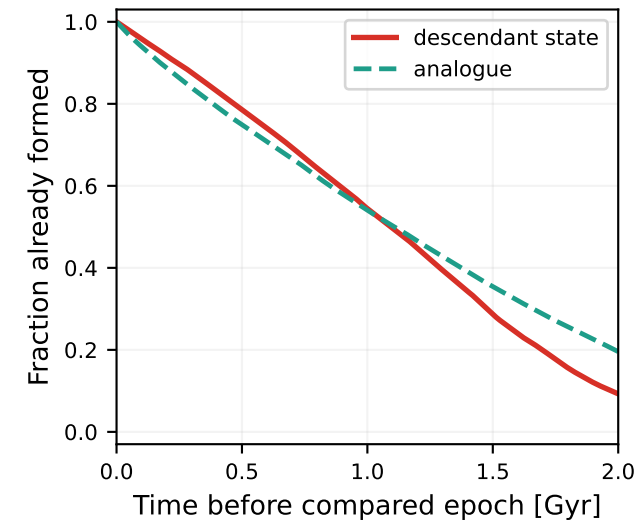
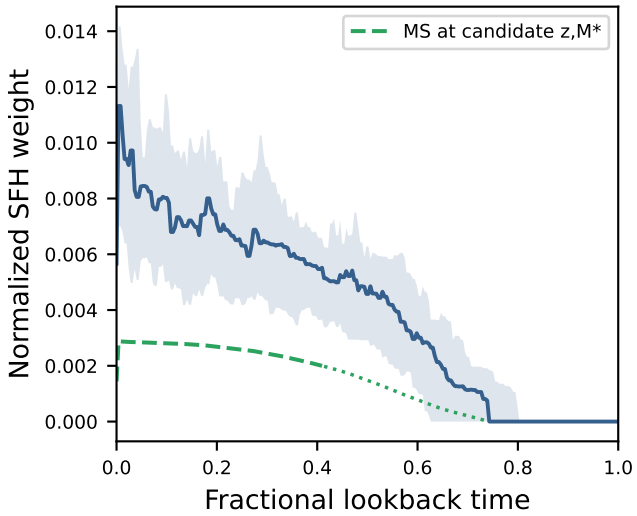


Earlier state $f=0.60$: lookback=3.23 Gyr, predicted $\log M^*=10.26$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

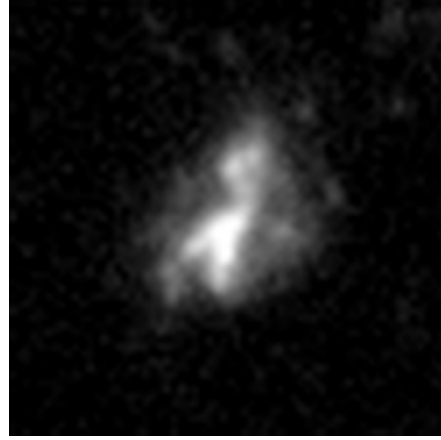
rank 1: 2640109785664925808
 $\log M^*=10.35$ (+0.09) $z=1.478$
 PHZ $\Delta MS=+1.75$; SFH-100Myr $\Delta MS=+0.56$
 $D=0.043$



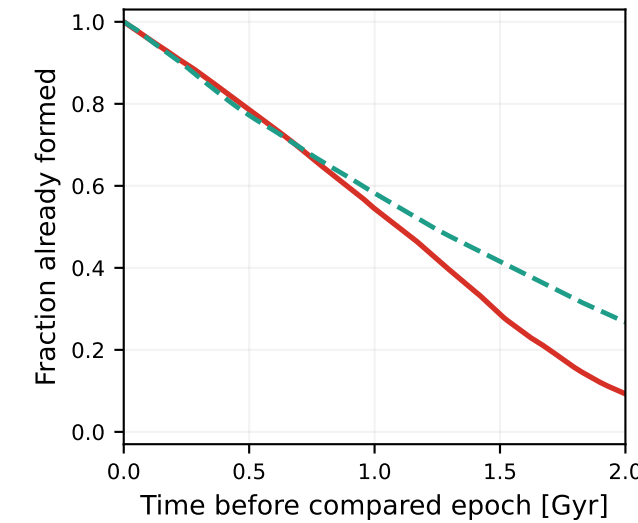
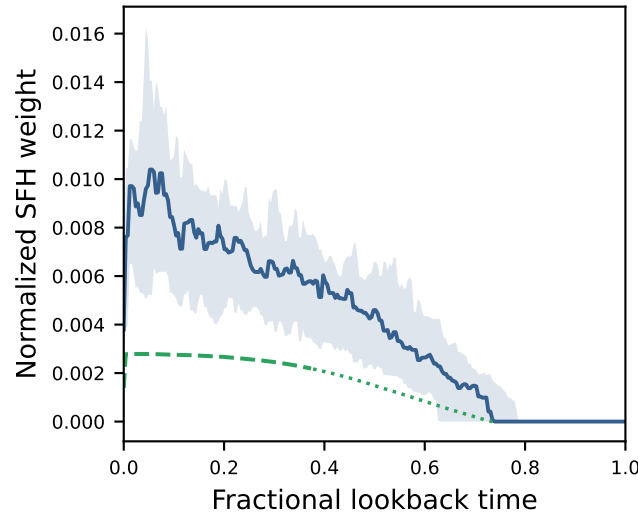
Complete candidate SFH



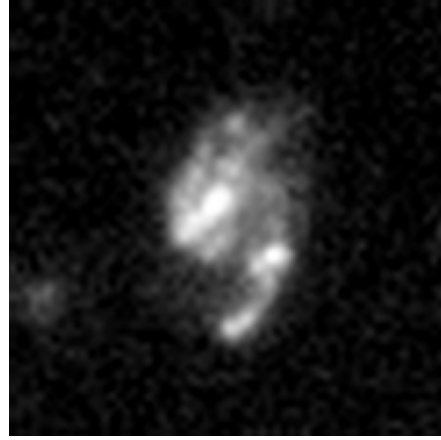
rank 2: 2717353722653902287
 $\log M^*=10.23$ (-0.03) $z=1.198$
 PHZ $\Delta MS=+1.81$; SFH-100Myr $\Delta MS=+0.49$
 $D=0.064$



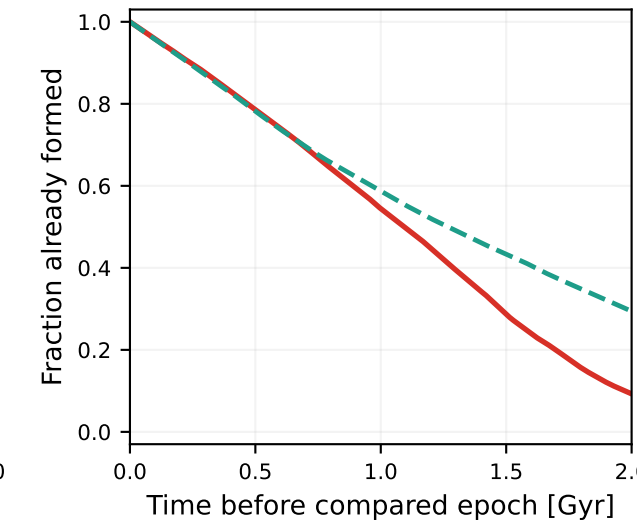
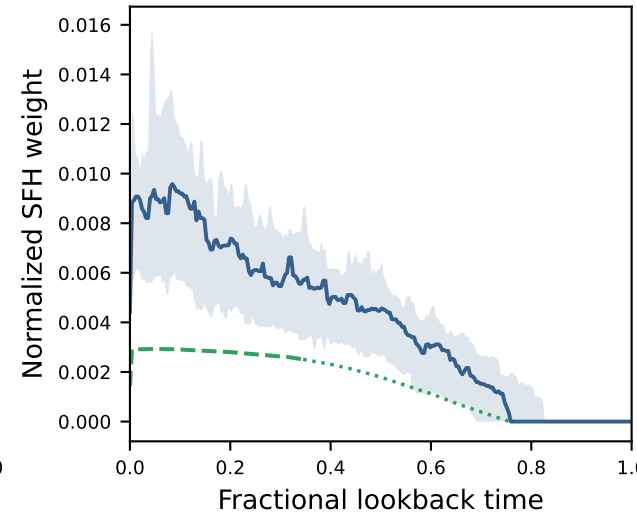
Complete candidate SFH



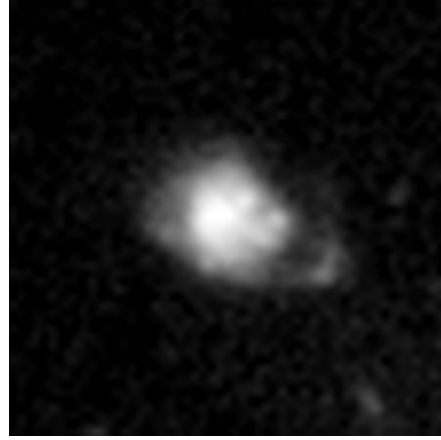
rank 3: 2691025900648883627
 $\log M^*=10.15$ (-0.12) $z=1.182$
 PHZ $\Delta MS=+1.66$; SFH-100Myr $\Delta MS=+0.49$
 $D=0.072$



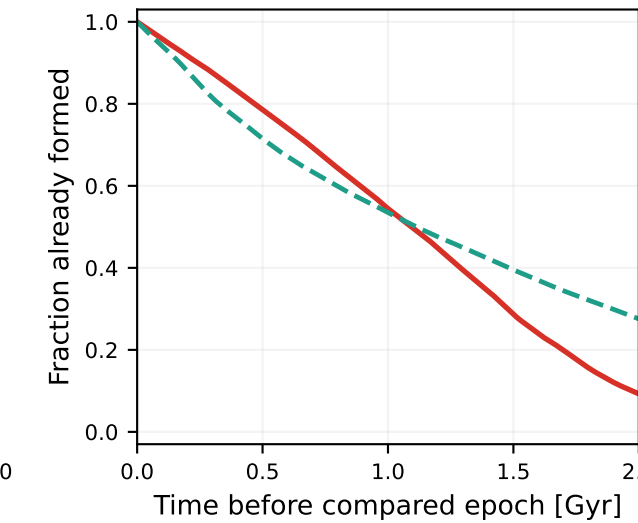
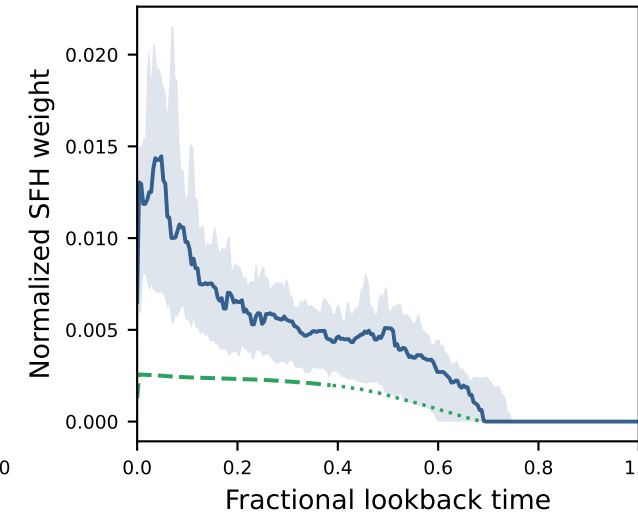
Complete candidate SFH



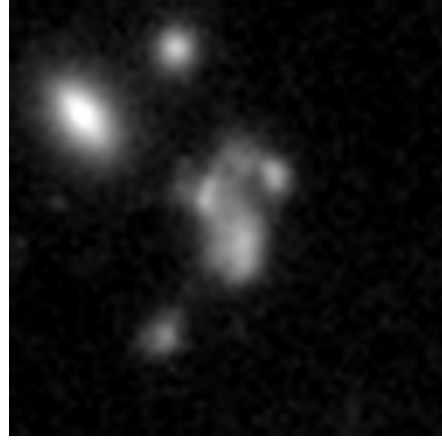
rank 4: 2656685702653947484
 $\log M^*=10.28$ (+0.02) $z=1.122$
 PHZ $\Delta MS=+1.26$; SFH-100Myr $\Delta MS=+0.69$
 $D=0.073$



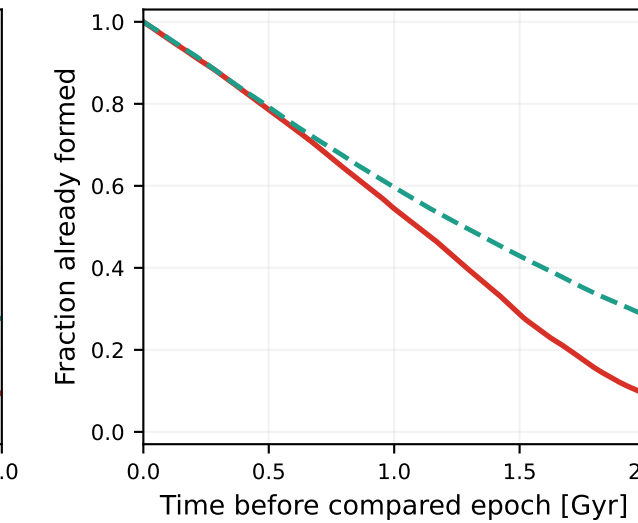
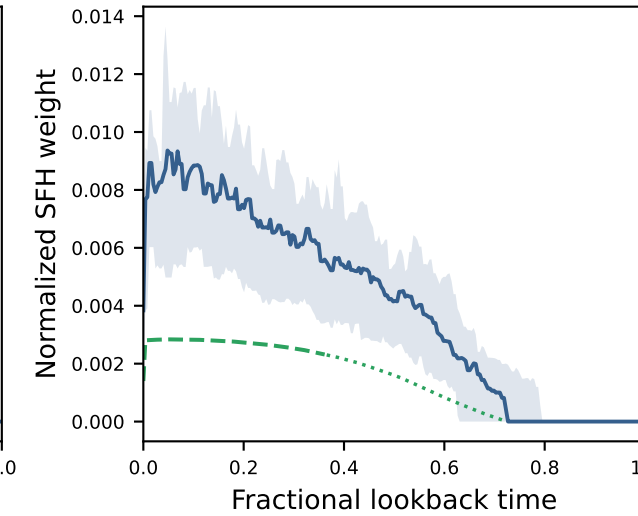
Complete candidate SFH



rank 5: 2737101536672032269
 $\log M^*=10.20$ (-0.07) $z=1.174$
 PHZ $\Delta MS=+1.67$; SFH-100Myr $\Delta MS=+0.69$
 $D=0.073$

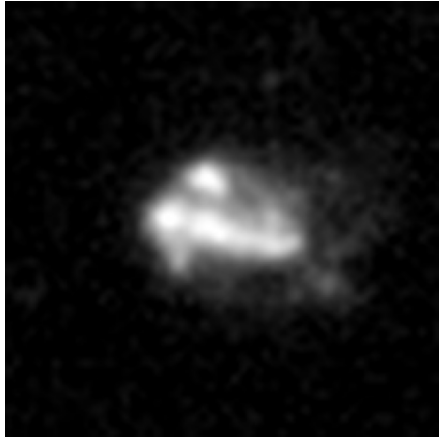


Complete candidate SFH

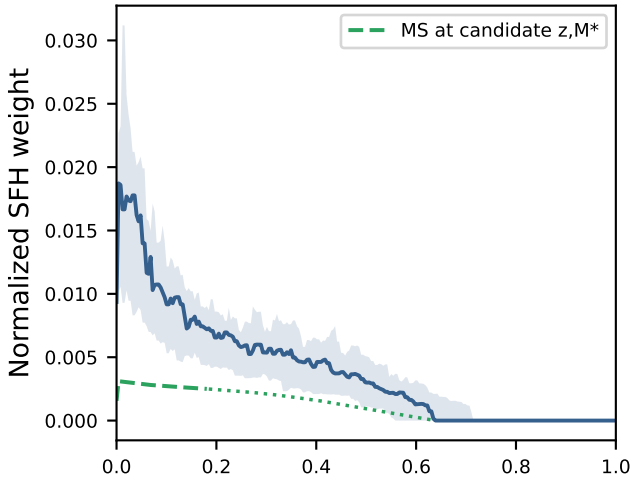


Earlier state $f=0.40$: lookback=3.99 Gyr, predicted $\log M^*=10.09$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

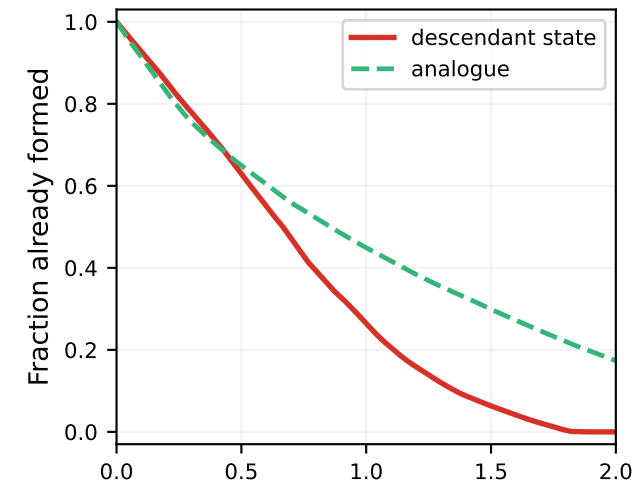
rank 1: 2647131718660160391
 $\log M^*=10.03$ (-0.06) $z=1.170$
 PHZ $\Delta MS=+1.63$; SFH-100Myr $\Delta MS=+0.76$
 $D=0.140$



Complete candidate SFH

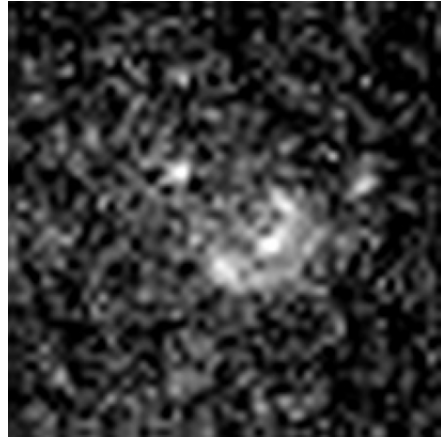


Fractional lookback time

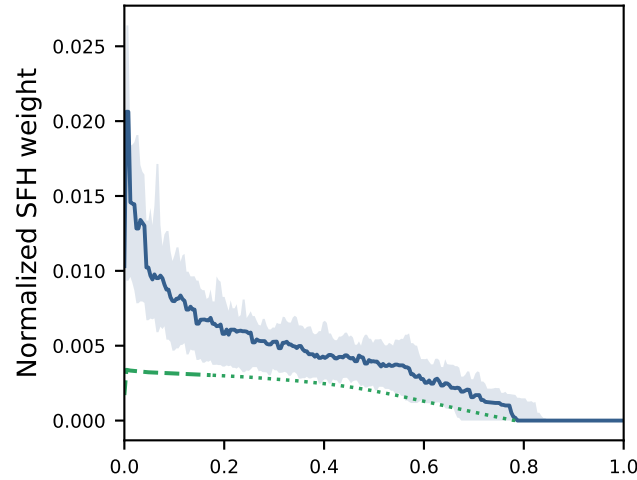


Time before compared epoch [Gyr]

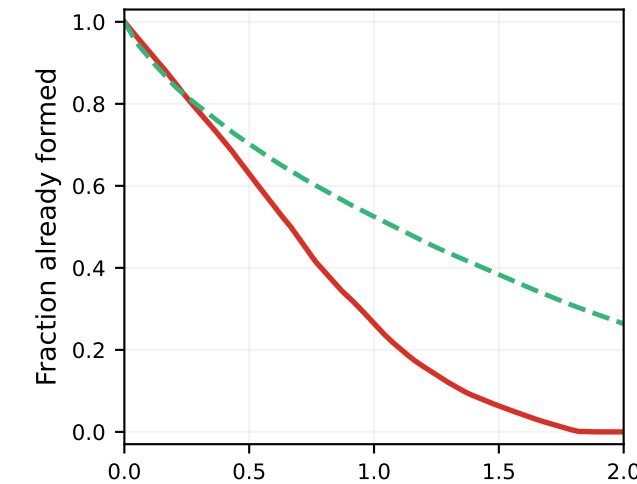
rank 2: 2647176063640000279
 $\log M^*=9.95$ (-0.14) $z=1.240$
 PHZ $\Delta MS=+1.62$; SFH-100Myr $\Delta MS=+0.72$
 $D=0.200$



Complete candidate SFH

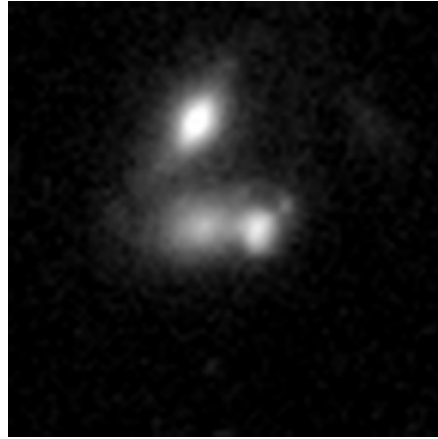


Fractional lookback time

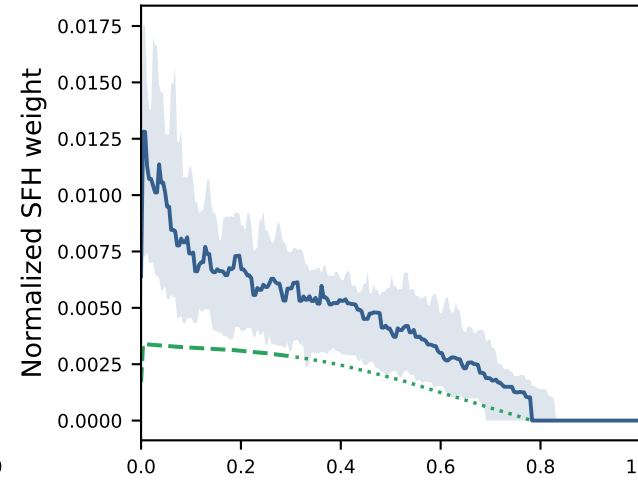


Time before compared epoch [Gyr]

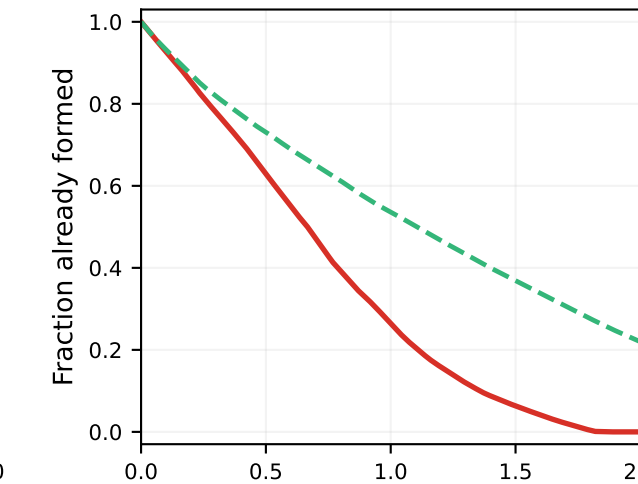
rank 3: 2658189408677769790
 $\log M^*=10.07$ (-0.01) $z=1.464$
 PHZ $\Delta MS=+1.59$; SFH-100Myr $\Delta MS=+0.54$
 $D=0.201$



Complete candidate SFH

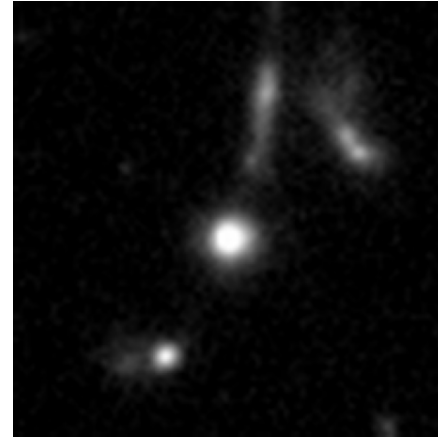


Fractional lookback time

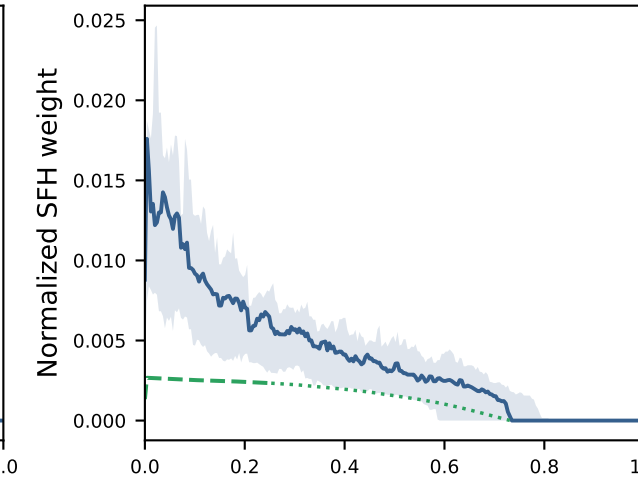


Time before compared epoch [Gyr]

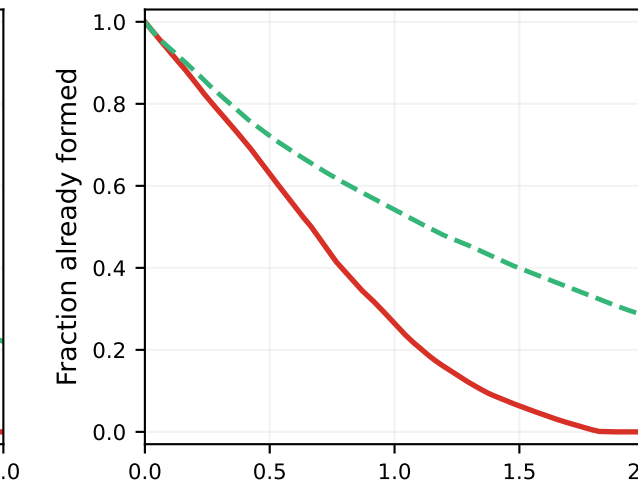
rank 4: 2684931694682266433
 $\log M^*=10.09$ (+0.00) $z=0.984$
 PHZ $\Delta MS=+1.06$; SFH-100Myr $\Delta MS=+0.76$
 $D=0.217$



Complete candidate SFH

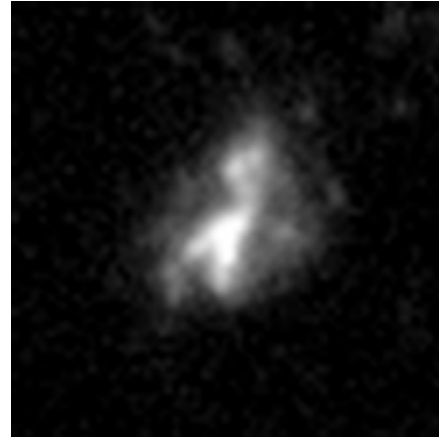


Fractional lookback time

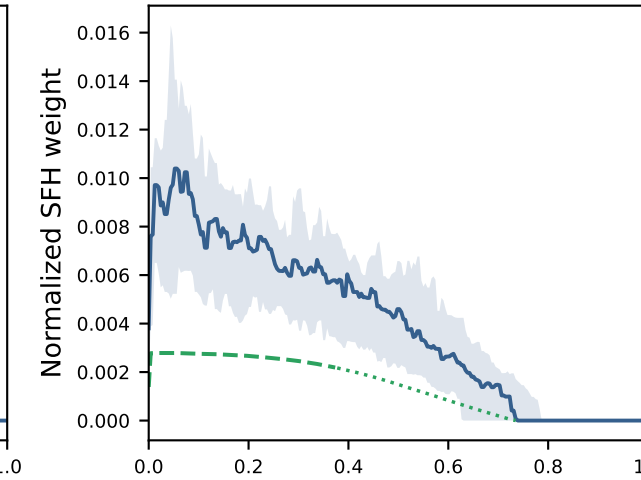


Time before compared epoch [Gyr]

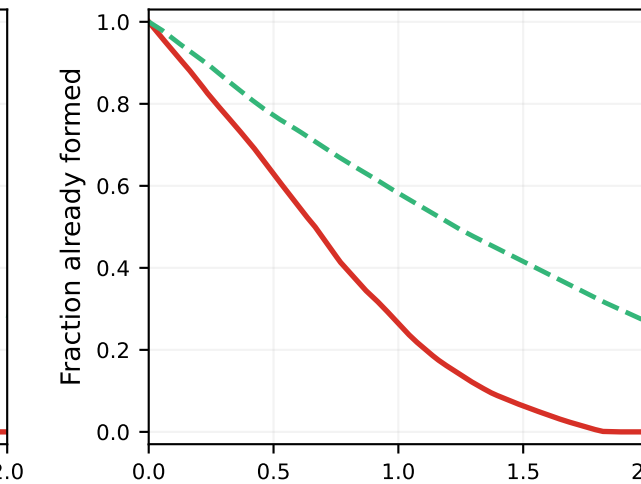
rank 5: 2717353722653902287
 $\log M^*=10.23$ (+0.14) $z=1.198$
 PHZ $\Delta MS=+1.81$; SFH-100Myr $\Delta MS=+0.0$
 $D=0.245$



Complete candidate SFH



Fractional lookback time



Time before compared epoch [Gyr]

Earlier state $f=0.20$: lookback=4.65 Gyr, predicted $\log M^*=9.79$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

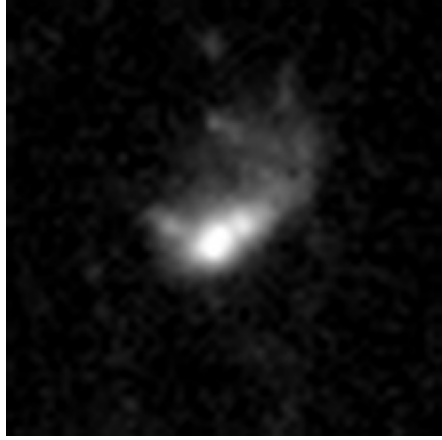
rank 1: 2645606478643808588
 $\log M^*=9.72$ (-0.06) $z=1.116$
 PHZ $\Delta MS=+1.67$; SFH-100Myr $\Delta MS=+0.83$
 $D=0.269$

rank 2: 2744318378668883710
 $\log M^*=9.65$ (-0.14) $z=0.770$
 PHZ $\Delta MS=+1.31$; SFH-100Myr $\Delta MS=+0.98$
 $D=0.284$

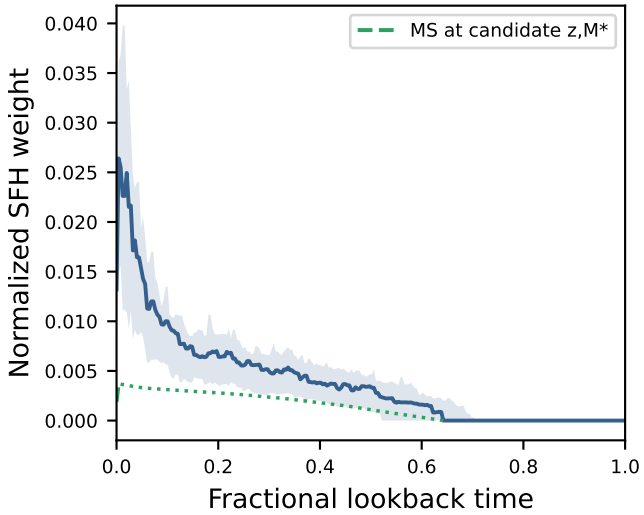
rank 3: 2766281064676170516
 $\log M^*=9.85$ (+0.06) $z=1.138$
 PHZ $\Delta MS=+1.74$; SFH-100Myr $\Delta MS=+0.72$
 $D=0.309$

rank 4: 2733550546657853038
 $\log M^*=9.76$ (-0.03) $z=0.760$
 PHZ $\Delta MS=+1.25$; SFH-100Myr $\Delta MS=+0.97$
 $D=0.313$

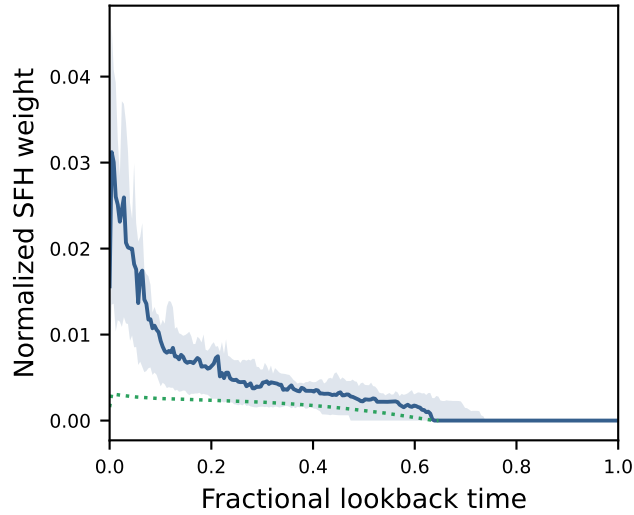
rank 5: 2650234495658951302
 $\log M^*=9.93$ (+0.14) $z=1.010$
 PHZ $\Delta MS=+1.36$; SFH-100Myr $\Delta MS=+0.97$
 $D=0.338$



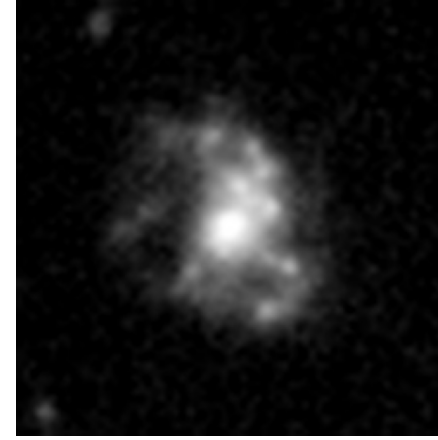
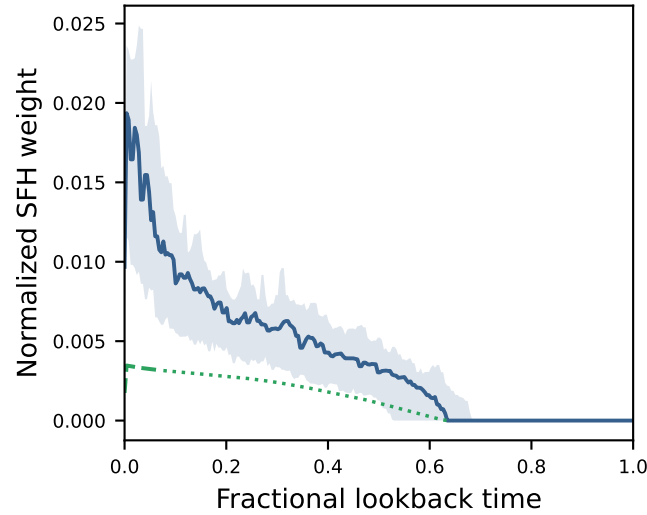
Complete candidate SFH



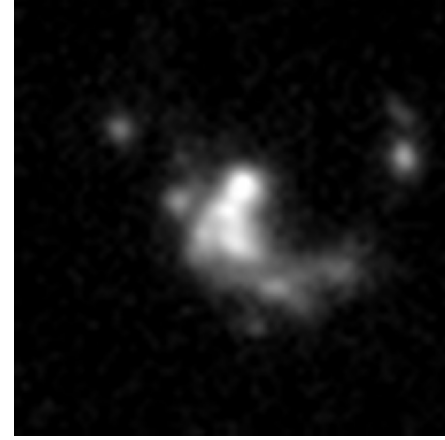
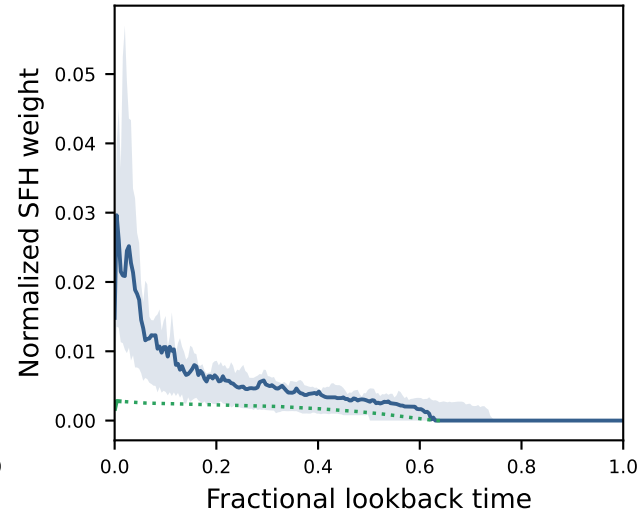
Complete candidate SFH



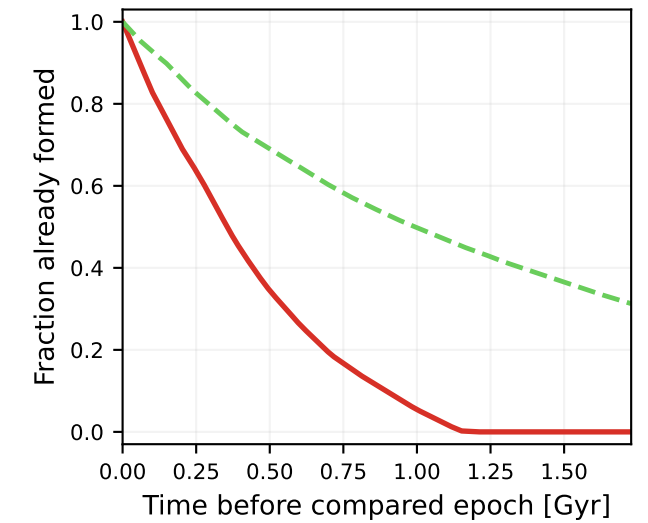
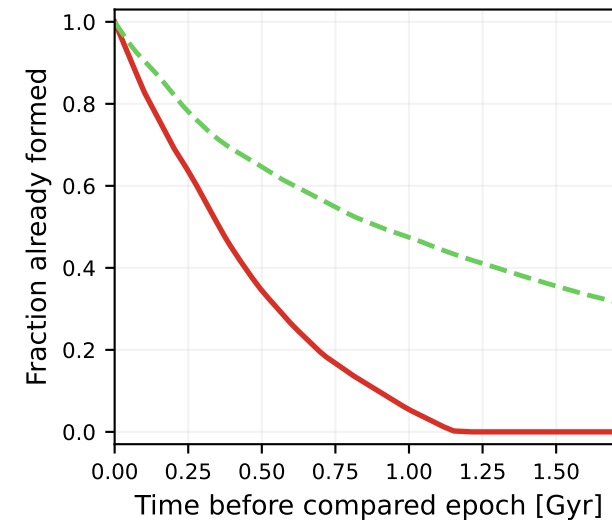
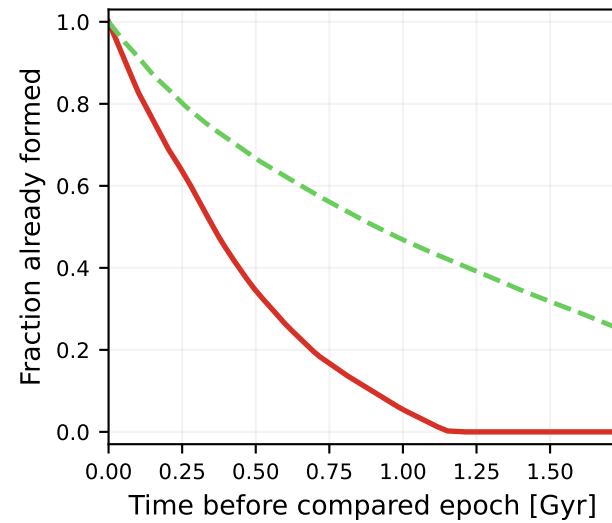
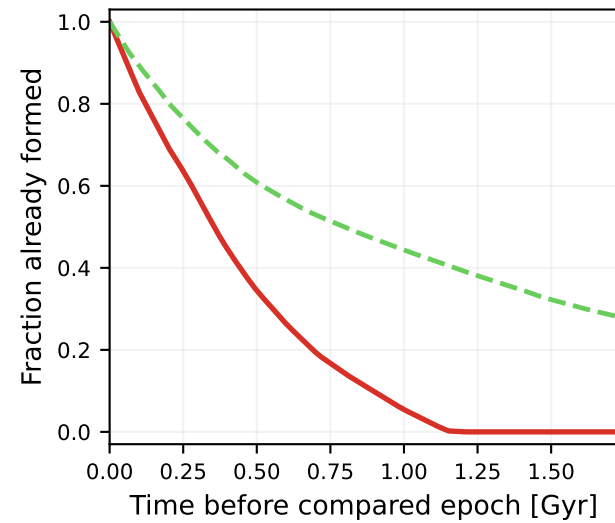
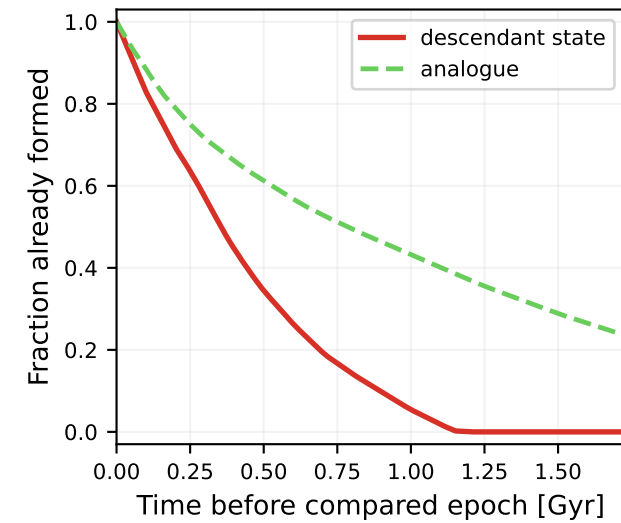
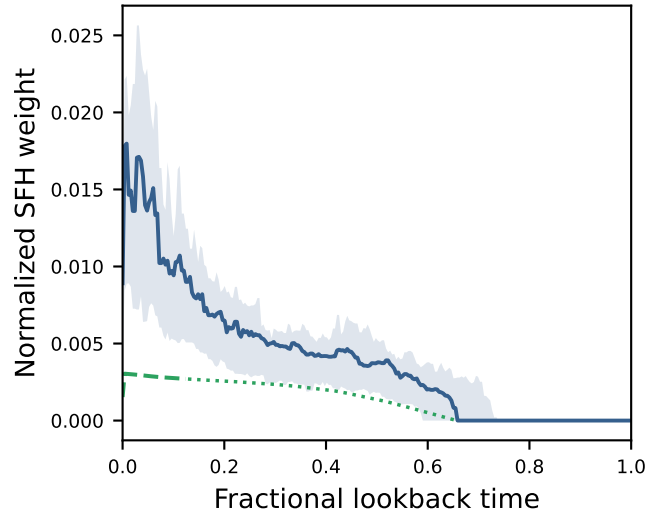
Complete candidate SFH



Complete candidate SFH



Complete candidate SFH

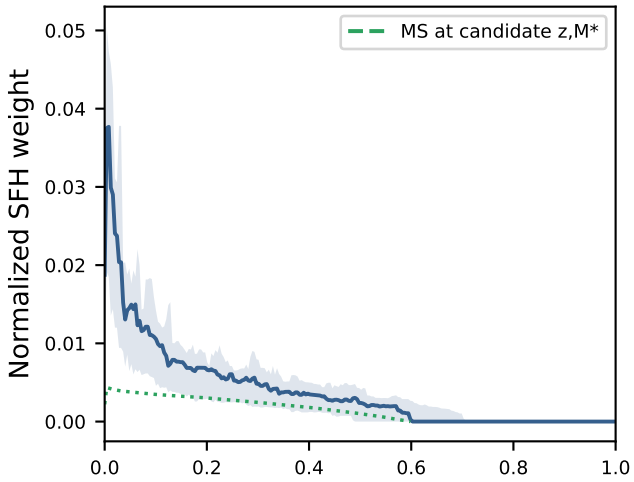


Earlier state $f=0.10$: lookback=5.01 Gyr, predicted $\log M^*=9.48$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

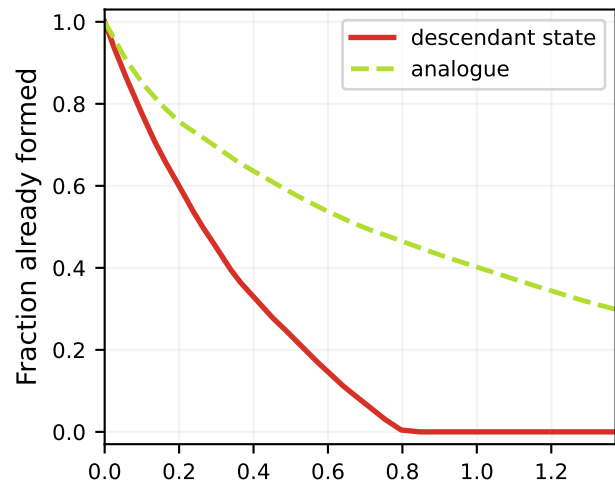
rank 1: 2648545772681764910
 $\log M^*=9.39$ (-0.09) $z=0.994$
 PHZ $\Delta MS=+1.79$; SFH-100Myr $\Delta MS=+0.90$
 $D=0.317$



Complete candidate SFH

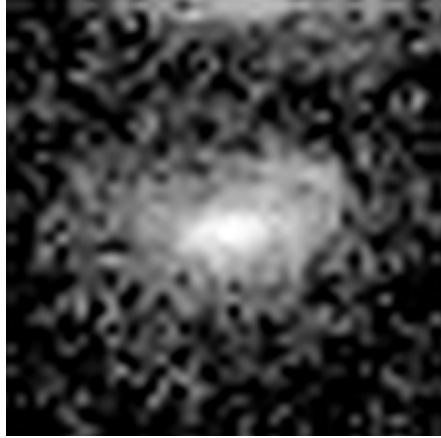


Fractional lookback time

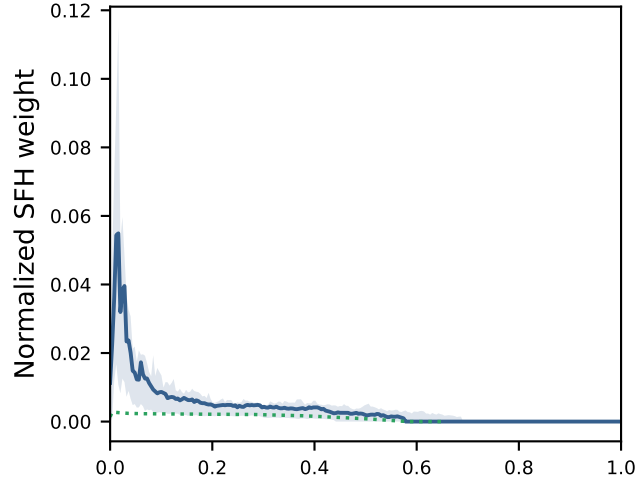


Time before compared epoch [Gyr]

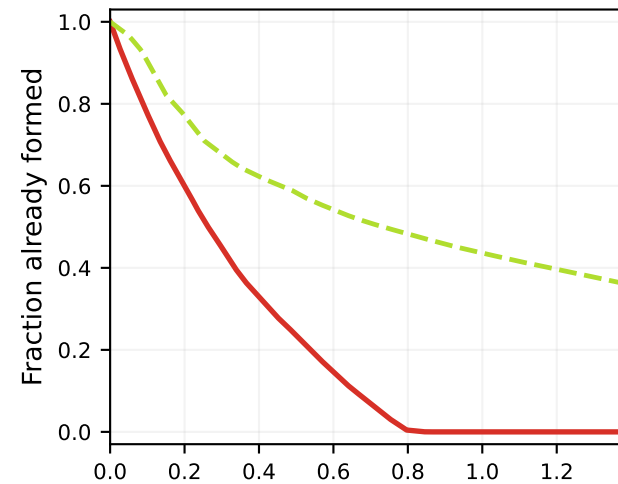
rank 2: 2647975483685285807
 $\log M^*=9.44$ (-0.05) $z=0.510$
 PHZ $\Delta MS=+1.34$; SFH-100Myr $\Delta MS=+1.07$
 $D=0.339$



Complete candidate SFH

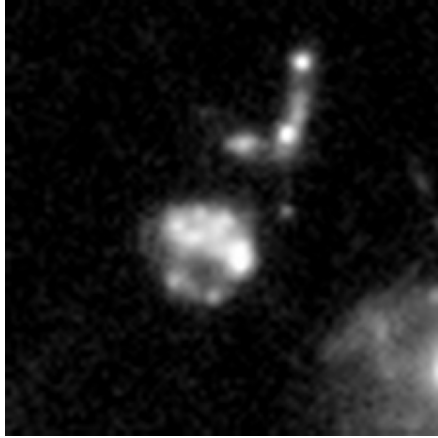


Fractional lookback time

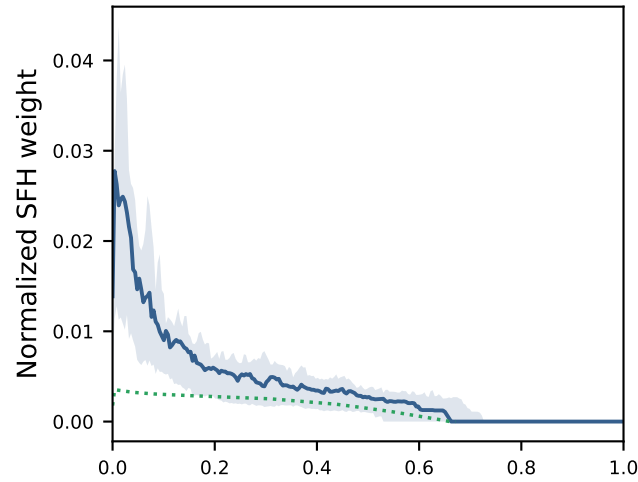


Time before compared epoch [Gyr]

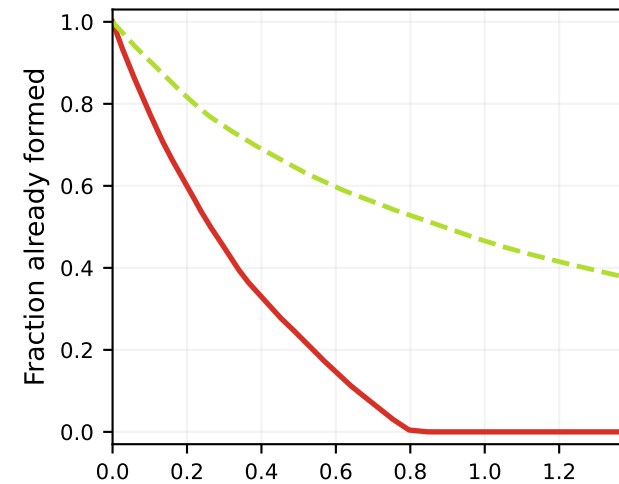
rank 3: 2740268308665100250
 $\log M^*=9.47$ (-0.02) $z=0.760$
 PHZ $\Delta MS=+1.77$; SFH-100Myr $\Delta MS=+0.87$
 $D=0.377$



Complete candidate SFH

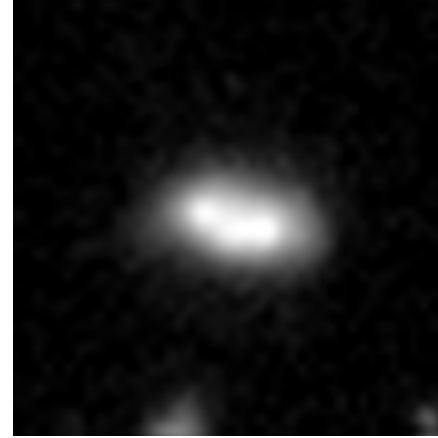


Fractional lookback time

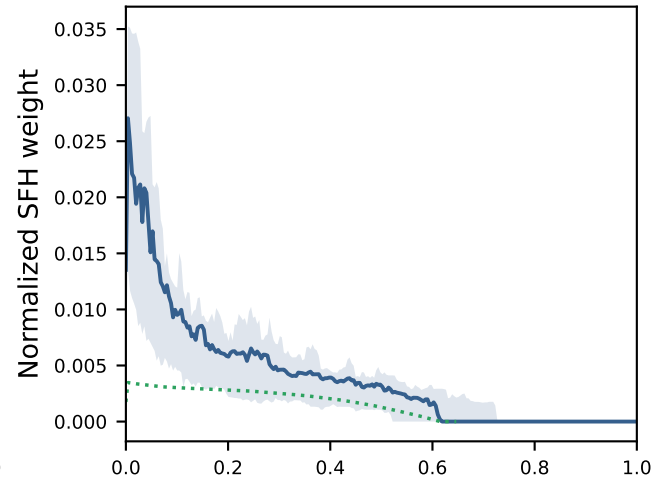


Time before compared epoch [Gyr]

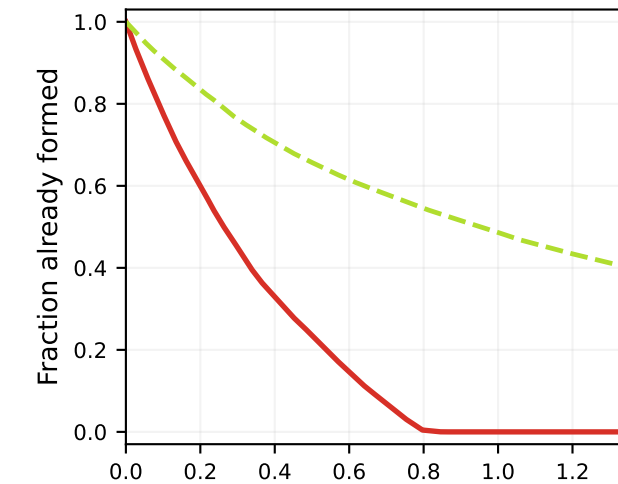
rank 4: 2729264667640595011
 $\log M^*=9.49$ (+0.00) $z=0.760$
 PHZ $\Delta MS=+1.78$; SFH-100Myr $\Delta MS=+0.86$
 $D=0.393$



Complete candidate SFH

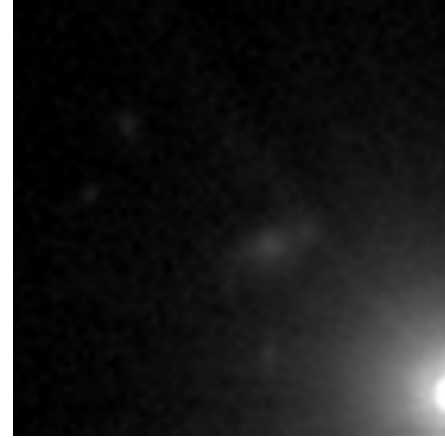


Fractional lookback time

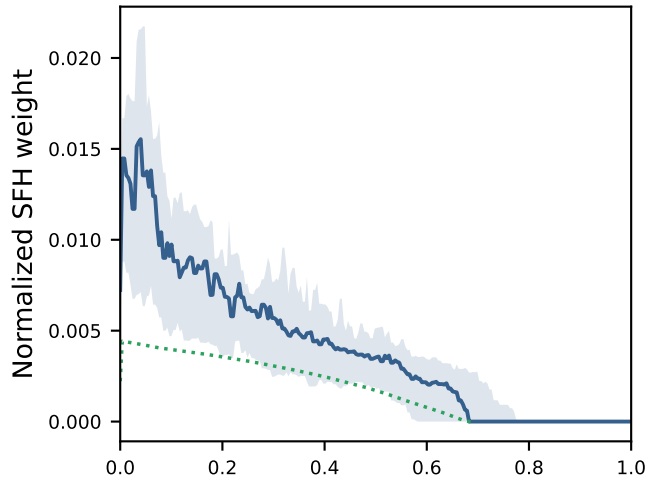


Time before compared epoch [Gyr]

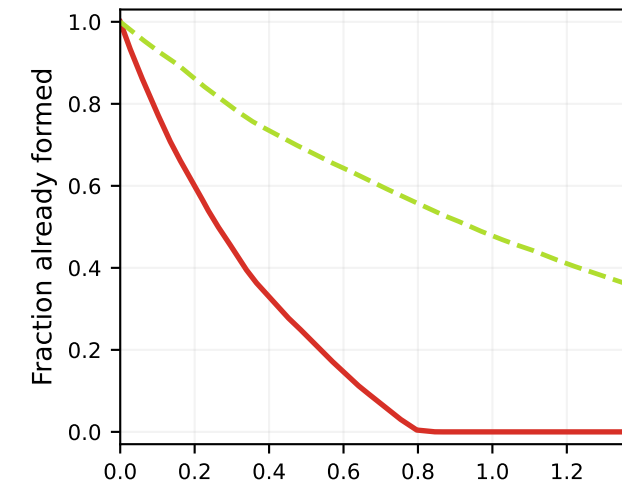
rank 5: 2695901997638798783
 $\log M^*=9.55$ (+0.06) $z=1.234$
 PHZ $\Delta MS=+1.48$; SFH-100Myr $\Delta MS=+0.5$
 $D=0.401$



Complete candidate SFH



Fractional lookback time



Time before compared epoch [Gyr]